

Complement-Mediated Kidney Diseases

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CHAPTER OUTLINE

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KEY POINTS

- In several diseases the tight homeostasis of the complement system is disrupted and results in overactivation of the complement system by genetic alterations, autoantibodies, or immune-complex formation.
- Atypical hemolytic uremic syndrome (aHUS) accounts for approximately 5% of all patients presenting with clinical hemolytic uremic syndrome. Factors that increase the likelihood of aHUS include presentation before ~12 months of age, positive family history of hemolytic uremic syndrome (in up to 30% of aHUS patients), and history of consanguinity.
- A pathogenic genotype is identified in 40% to 60% of patients with aHUS.
- Anti-CFH autoantibodies account for 10% to 25% of aHUS cases (up to 55% in India).
- The current gold standard therapy for aHUS is with C5 inhibitors; however, in less-well resourced countries plasma exchange or plasma infusions are often the only available or accessible treatment options.
- C3 glomerulopathy (C3G) and immune-complex membranoproliferative glomerulonephritis are characterized by proteinuria, hypertension and impaired kidney function, and patients are at significant risk of progression to kidney failure.
- Therapy includes steroids and mycophenolate mofetil, although C3-targeting therapies show promising results in trials and might become first-line therapy soon.

THE COMPLEMENT SYSTEM

The complement cascade consists of plasma proteins (C1–C9) and either surface-bound or fluid-phase (plasma) regulatory proteins that prevent harmful complement activation on host cells like endothelial cells (Fig. 36.1). The stages of complement activation can be divided into initiation, amplification, and termination. The classical pathway (CP) is activated when C1q binds to antigen-antibody complexes and the lectin pathway by contact with microbial carbohydrates. In contrast, the alternative pathway (AP) lacks a specific activation trigger; therefore it is constitutively active to permit a rapid response to pathogen or damage-associated signals. Furthermore, the alternative pathway provides the

substrates for all complement pathways as they converge on the C3 convertase. The alternative pathway ticks over through spontaneous hydrolysis of C3, which produces C3a—an anaphylatoxin/messenger molecule—as well as C3b. C3b, in conjunction with factor D, facilitates the cleavage of factor B to Bb. C3b and Bb complexes bind and form the alternative pathway C3 convertase, C3bBb. Both the classical and lectin pathway C3 convertase (C2aC4b) and alternative pathway C3 convertase cleave C3 into C3a and C3b. C3b can then either enter the amplification loop and generate more C3 convertases (it can bind to foreign but in the same way also to host surfaces and lead to opsonization) or a small part moves forward to the terminal pathway by generating a C5 convertase (C3bBbC3b). The C5 convertase cleaves C5 into C5a, another potent anaphylatoxin/messenger molecule,

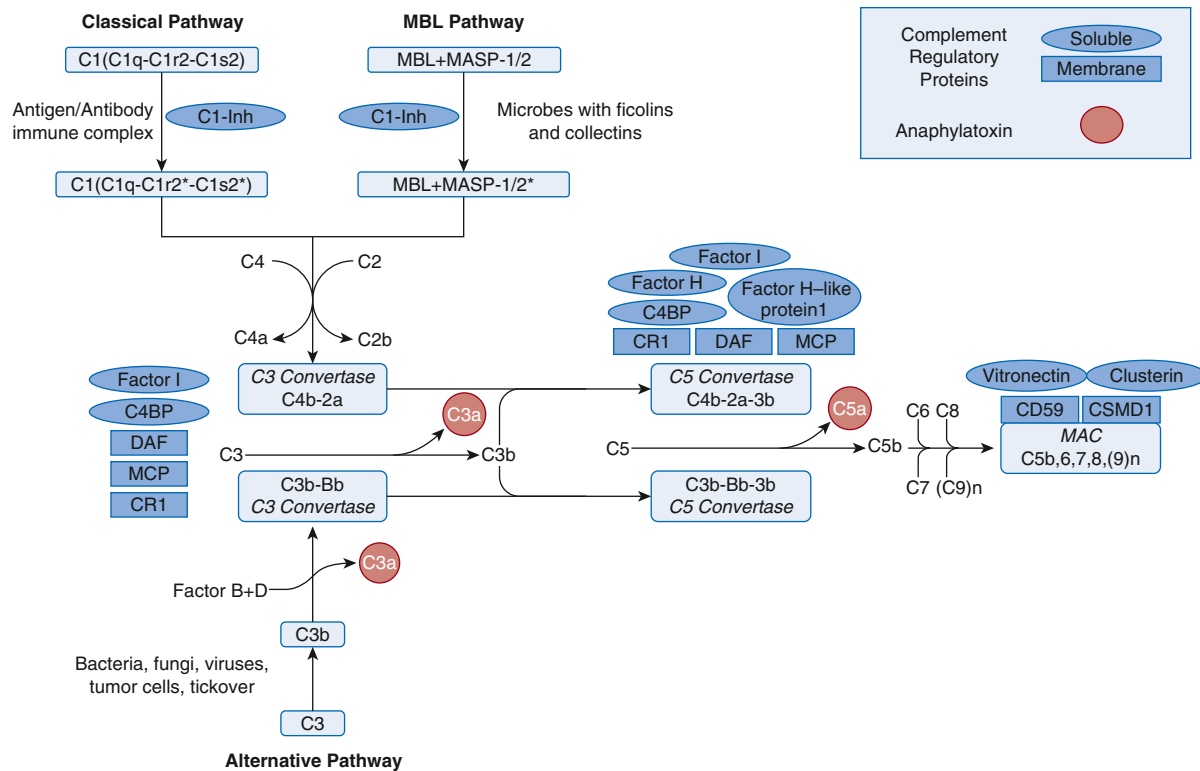


Fig. 36.1 The complement system and its alternative pathway. The alternative pathway of complement (AP) begins with spontaneous hydrolysis of C3 and the production of C3b. C3b combines with Bb to form the C3 convertase. The C3 convertase may go on to form the C5 convertase (C3bBbC3b) after the addition of another C3b or it may feedback to make additional C3b in an amplification loop. C5 convertase will cleave C5, producing C5b, the first protein in the terminal complement complex. The ultimate products of the AP include two important anaphylatoxins (C3a and C5a) and the C5b-9 complex. Factor H is the major negative regulator of the AP, and dysregulation of FH is the most common genetic cause of aHUS. (Created in BioRender. Licht C. 2025. <https://BioRender.com/0w57eov>.)

and C5b, which can bind C6, C7, C8, and multiple C9 molecules to form the terminal complement complex C5b-9 or membrane attack complex (MAC).¹ This complex can be fully inserted into the plasma membrane of cells, form a pore, and result in cell injury and death.²

To prevent overactivation of the complement system and insertion of the C5b-9 pore into host cells, biological systems have developed several layers of complement regulation. The regulators mainly modify the amplification loop and C3 convertase function and occur either in the fluid phase (complement factor H, complement factor H-like proteins) or on the surface of host cells (CD46/membrane cofactor protein [MCP], CD55, CD59, C1R). These regulators facilitate the dissociation of C3 and C5 convertase or act as cofactors for the factor I-mediated decay of C3b and C4b. Complement factor H (CFH) is a soluble complement regulator produced in the liver. It can protect from unwanted complement activation on host cells through its ability to bind to glycosaminoglycans and sialic acids present only on host cells and is absent in most pathogens. The binding then allows for C3b inactivation on host cells. Mutations of the C-terminus of CFH, which interfere with the binding to host cells, have been associated with atypical hemolytic uremic syndrome (aHUS), whereas N-terminal mutations, which affect the cofactor and decay function, have been seen in patients with C3 glomerulopathy (C3G).¹ CD59 prevents the insertion of C5b-9, so it acts on the terminal pathway.

In several diseases this tight homeostasis is disrupted and results in overactivation of the complement system^{1,3-5} by the following means:

1. Genetic alterations that result in loss of function mutations of regulators or gain of function mutations of complement proteins (e.g., aHUS, C3G)
2. Autoantibodies that bind negative regulators and inhibit their function or stabilize convertases, resulting in continuous activation of the cascade (e.g., aHUS, C3G)
3. Immune-complex formation, which activates the classical pathway (e.g., acute poststreptococcal glomerulonephritis and systemic lupus erythematosus [SLE])
4. Overactivation of the complement cascade in general (e.g., secondary forms of thrombotic microangiopathy [TMA]), which are most likely multifactorial (e.g., endothelial cell activation, inflammation, prothrombotic state, heme deposition)

Overactivation of the complement system then results in:

1. Deposition of complement products in the kidney tissue as seen in C3G, SLE, acute poststreptococcal glomerulonephritis, and/or
2. C5b-9 formation on endothelial cells resulting in endothelial cell activation and release of prothrombotic molecules, inflammation, thrombi formation, and the clinical picture of TMA, and/or

3. Inflammation via anaphylatoxins (C3a/C5a) and neutrophil recruitment and activation as can be seen in all of the previously mentioned diseases.

Here we discuss atypical hemolytic uremic syndrome (aHUS), a thrombotic microangiopathy caused by complement mutations and autoantibodies resulting in the triad of hemolytic anemia, thrombocytopenia, and kidney disease. We also address C3G and immune complex membranoproliferative glomerulonephritis (IC-MPGN), two kidney diseases secondary to complement deposition within the mesangium of the glomerulus leading to inflammation and glomerular dysfunction mainly secondary to autoantibodies that stabilize C3, C4, or C5 convertase complexes but which can also result from genetic mutations of complement proteins. We also discuss the broader TMA spectrum, which includes several secondary forms, where the role of the complement system can be evident but is less well established. Acute poststreptococcal glomerulonephritis is the most common cause of a nephritic syndrome presentation in children and is discussed as the prototype of a postinfectious immune complex-mediated GN and an important differential diagnosis for C3G/IC-MPGN.

THROMBOTIC MICROANGIOPATHIES

ATYPICAL HEMOLYTIC UREMIC SYNDROME

EPIDEMIOLOGY

Atypical HUS is a rare disease. Most experts agree that aHUS accounts for approximately 5% of all patients presenting with the clinical findings of HUS, but this depends on the geographic area. Argentina, with its high cow meat intake, has a high rate of Shiga toxin-producing *Escherichia coli* (STEC) HUS, whereas in India 50% of HUS patients have CFH antibodies.^{6,7} Other studies refer to a prevalence of 2 and 7 cases per 1 million population in adults or children, respectively.⁸⁻¹⁰ In children, aHUS affects females and males equally, while in adults it occurs more frequently in females.¹¹ A few factors increase the likelihood that a patient with a new clinical diagnosis of HUS will ultimately be confirmed to have aHUS: presentation before ~12 months of age,^{12,13} positive family history of HUS (in up to 30% of aHUS patients),¹⁴ and history of consanguinity.

CLINICAL FEATURES

Patients with aHUS often present precipitously, often on the background of constitutional symptoms, such as fatigue and anorexia. A preceding illness is often reported, including a gastrointestinal infection (causing nonbloody diarrhea) or a respiratory illness.¹⁵⁻²¹

The classic diagnostic triad of TMA is usually evident on initial presentation. This includes microangiopathic hemolytic anemia (often presenting as pallor), thrombocytopenia or decrease in platelet counts (presenting as purpuric rash, easy bruising, or mucosal bleeds such as gingival bleeding), and target organ damage (namely acute kidney injury [AKI], which may present as hypertension, decreased urine output, edema, etc.).^{14,15} It is important to note that 10% to 20% of patients might not present with thrombocytopenia and only show evidence of

microangiopathic hemolysis and AKI. Clinical outcomes in these patients are similar to those of patients with “full aHUS,” so a high clinical suspicion is warranted even in the absence of a typical presentation.²²

DIAGNOSIS

The diagnosis of aHUS is clinical and a diagnosis of exclusion. After a TMA is diagnosed, it is important to investigate the underlying etiology. In children, the first diagnosis to consider is STEC HUS, which is the most common form of TMA in this age group and often has a preceding gastroenteritis, commonly with bloody diarrhea (discussed further in [Chapter 71](#)). To exclude STEC HUS, stool for Shigatoxin should be sent to the laboratory. Next, thrombotic thrombocytopenic purpura (TTP) needs to be excluded (especially in adolescents and adults) by measuring ADAMTS13 activity. It is important to note, however, that reduced ADAMTS13 activity (but with levels remaining >10%) is frequently seen in aHUS and secondary forms of HUS due to endothelial cell activation. If the activity is <10%, ADAMTS13 antibodies should be measured.²³ Importantly, samples for ADAMTS13 testing must be sent before the start of any plasma exchange or plasma infusion. In all patients with HUS/TMA, planning for all eventual required blood testing before initiation of testing, with correct handling and storage of samples, is key to permit the stepwise approach to a correct diagnosis.

In children and especially adults, other secondary forms of TMA should also be excluded, such as infections (*Streptococcus pneumoniae*, H1N1, COVID-19), multiple drugs, solid organ transplantation, bone marrow/stem cell transplantation, and other autoimmune or kidney diseases such as SLE or C3G/MPGN.⁴ HUS occurring in association with infections in adults, especially during pregnancy, may be difficult to interpret as these conditions may cause HUS alone or may be triggers for aHUS, leading to diagnostic confusion and delays in implementing effective therapies.²⁴

While the kidneys are the primary target in all forms of aHUS, extrarenal involvement is present in 20% to 50% of patients.²⁵ The range of extrarenal organs affected in patients with aHUS closely overlaps with that reported for patients with STEC HUS.²⁶ The most frequent extrarenal symptom is seizures, but other neurologic abnormalities may also be observed including diplopia, blindness, paresis, and coma. Prompt neuroimaging is crucial in discerning whether these events are due to endothelial cell injury of the CNS vasculature or a result of underlying kidney pathology (e.g., posterior reversible encephalopathy syndrome [PRES]).²⁷ Cardiac dysfunction including pericardial effusion or myocardial infarction has been reported.²⁸ Gastrointestinal involvement may occur in the form of hepatitis, pancreatic necrosis, and/or ischemic colitis. Vasculopathies of peripheral limbs or the retina are rare.²⁵

In terms of laboratory investigations, circulating C3 and C4 complement assays are not useful since only 30% to 40% of patients with aHUS will have abnormal levels^{15,16} and normal serum C3 levels do not rule out aHUS. Even mutations in *CFH* or *CFI* may be associated with normal plasma levels.¹⁵ The presence of elevated soluble C5b-9 levels in serum may be a useful indicator of disease activity and terminal pathway activation, but it is costly and not widely available and its predictive value has not been investigated.²⁹

While genetic testing with current aHUS gene panels is invaluable to distinguish aHUS subtypes, a detailed report is generally only available after a few weeks, too late for clinical decision making in the acute setting. Furthermore, in ~40% to 60% of cases, a significant genetic association may not be detected.^{14,30,31} Genetic studies, however, are important to guide subsequent treatment and are useful for long-term management. Current aHUS genetic testing panels include *CFH*, *CFI*, *MCP*, *C3*, *CFB*, and *diacylglycerol kinase epsilon* (*DGKE*). Whole-exome sequencing (WES) or whole-genome sequencing (WGS) can aid in patients where the genetic diagnosis is unclear.³² The detection of hybrid *CFH* genes and copy-number variations in *CFH* or *CFHR1–5* is done using multiplex ligation-dependent probe amplification (MLPA).³³ Investigations for anti-*CFH* antibodies should also be sent immediately after diagnosis and before treatment initiation. Patients with findings consistent with cobalamin C deficiency should undergo detailed biochemical characterization and genetic testing for metabolism of cobalamin-associated C (*MMACHC*) mutations.³⁴

PATHOPHYSIOLOGY

Atypical HUS is a TMA, a histologic term that describes clot formation in capillaries of the kidney. The cause of the TMA cannot be distinguished by biopsy. It remains unclear why such a diverse array of HUS subtypes all exhibit exquisite tropism for the kidney microvasculature. Besides the architectural characteristics of the fenestrated endothelium of the glomerular capillaries, this may at least be in part related to their unique rheologic properties, as they represent the only capillary system in the body that is under high flow and high pressure.³⁵

TMA lesions are initiated when a severe injury to glomerular capillaries of multiple vascular beds triggers repair mechanisms that lead to a local prothrombotic environment. The microthrombi formed in aHUS trap platelets and lead to the observed thrombocytopenia. Shearing of erythrocytes as they transit through partially obstructed glomerular capillaries results in microangiopathic hemolytic anemia. The initiating factor in most patients with aHUS is related to abnormal activity of the alternative pathway of the complement system (see Fig. 36.1), which leads to C5b-9 activation on glomerular endothelial cells and subsequent injury.^{2,36} The pathophysiology of TMAs related to infections, drugs, transplants, and other glomerular diseases are less well understood. However, it is of utmost importance to consider complement as a driving force in the disease pathogenesis because these patients might benefit from complement inhibition therapy.⁴

CAUSES OF aHUS

Most reviews on aHUS suggest that ~60% of patients have a pathogenic genotype.^{14,15} A study on more than 3100 aHUS patients from 6 major centers reported that only ~40% had a mutation in one of the major aHUS genes.³¹ Clinicians should consider including a genetic testing panel in their early management care plan as genetic testing results will help predict the overall prognosis and response to established therapies, determine the optimal timing of eculizumab discontinuation if used, may suggest other organ systems that might be at risk, and may aid in planning for suitable pretransplant and posttransplant precautions, as

well as family planning.²⁴ Next, we discuss the major forms of genetic aHUS associated with complement abnormalities, followed by genetic aHUS subtypes that cause pathology independent of the complement system and an autoimmune form of aHUS secondary to CFH-autoantibodies (genetics of aHUS is further discussed in Chapter 44).

aHUS has also been linked to other conditions, mainly conditions that result in endothelial cell injury and/or can amplify the complement cascade (Table 36.1). These conditions can lead to transient complement activation and aHUS or can serve as a trigger for an underlying genetic aHUS and hence genetic testing is recommended. Complement targeted therapy has also been used in many of these secondary forms. Although there are no recommendations for using complement targeting therapy in most of the conditions, as will be discussed later, there may certainly be a rationale for using it as a short-term therapy to calm down the complement activation and stop the TMA until the primary disorder can be brought under control.⁴

Genetic forms of complement-mediated aHUS

Pathogenic variants in a host of complement genes are often identified when investigating patients with aHUS (e.g., *CFH*,

Table 36.1 Overview of Complement Amplifying Conditions that can Associate With Thrombotic Microangiopathy (TMA)

Pregnancy-associated TMA (pTMA)
Post-hematopoietic stem cell transplant-associated TMA or transplant TMA (TA-TMA)
TMA post solid organ transplantation
TMA associated with metabolic disease: cobalamin C deficiency
TMA associated with systemic autoimmune diseases
• Systemic lupus erythematosus
• Antineutrophil cytoplasmic antibody-associated vasculitis
• Antiphospholipid syndrome
• Systemic sclerosis
• Mixed connective tissue disorder
• Sjögren syndrome
• Inflammatory bowel disease
TMA associated with drugs
• Quinine
• Anti-VEGF inhibitor
• Chemotherapy such as gemcitabine, mitomycin, and cisplatin
• Calcineurin inhibitors
• Gene therapy
TMA associated with malignant hypertension
TMA associated with infections
• Bacteria associated with a gastrointestinal infection, such as Shiga toxin producing <i>Escherichia coli</i> , <i>Shigella dysenteriae</i> , <i>Campylobacter jejuni</i> , <i>Moraxella osloensis</i>
• Viruses, such as cytomegalovirus, Epstein-Barr virus, varicella zoster virus, parvovirus B19, human immunodeficiency virus, influenza
• Bacteria causing respiratory tract infections, such as <i>Bordetella pertussis</i> , <i>Streptococcus pneumoniae</i> , <i>Mycoplasma pneumoniae</i> , COVID-19
• Fungal infections in immunocompromised hosts
TMA associated with other renal diseases
• C3G/MPGN
• IgA Nephropathy

CFI, *CFB*, *C3*, *MCP*). Genes encoding *negative regulators* of complement (*CFH*, *CFI*, *MCP*) are expected to harbor loss-of-function variants, while *positive regulators* (*C3*, *CFB*) are expected to have gain-of-function variants.⁴

The genetics of complement-driven aHUS is complex because it includes allelic heterogeneity (different variants in the same gene lead to similar phenotypes), locus heterogeneity (variants in different genes lead to similar phenotypes), and mode-of-inheritance heterogeneity (heterozygous or homozygous variants in the same gene can cause the same disease).³⁷ Importantly, three genes also exhibit phenotypic heterogeneity since variants in *CFH*, *C3*, or *CFHR5* can cause not only aHUS but also C3 glomerulopathy (a phenotypically distinct kidney disease; later). Interestingly, most of these conditions exhibit incomplete penetrance (a given variant is present in affected and nonaffected relatives) and/or variable expressivity (all relatives with a given variant are affected by the same disease with a wide spectrum of severity).³⁷

Complement factor H-associated aHUS. Complement factor H (*CFH*) is one of the major negative regulators of the alternative complement pathway: It regulates complement activation on host cells and surfaces by possessing both cofactor activity for factor I-mediated C3b cleavage, as well as activity for decay of the AP C3-convertase¹ (see Fig. 36.1). The discovery of mutations in *CFH* provided the first proof of the existence of a genetic form of aHUS.³⁸ Mutations in *CFH* are the most frequent pathogenic genotype identified in aHUS patients. They account for ~40% of aHUS cases with a confirmed molecular diagnosis³¹ (Fig. 36.2). Most mutations in *CFH* that lead to aHUS are associated with normal plasma *CFH* levels, but the protein is either nonfunctional or has reduced function.¹¹ Patients with homozygous *CFH* mutations typically present in infancy with severe disease, whereas patients with heterozygous mutations present at any age with milder forms of aHUS.³⁹

Membrane cofactor protein-associated aHUS. Membrane cofactor protein (*MCP*; CD46) is a cell-surface negative complement regulator. It functions as a cofactor for *CFI*, thereby assisting with C3b degradation¹ (see Fig. 36.1). The link between *MCP* mutations and aHUS was first established in 2003, with ~75% of kindreds exhibiting incomplete

penetrance in the context of a heterozygous mutation.^{40,41} More severe disease is associated with homozygous or compound heterozygous loss of function *MCP* mutations or heterozygous *MCP* mutations accompanied by another mutation in one of the complement genes.⁴² Pathogenic mutations in *MCP* are reported in ~16% of aHUS patients with a confirmed molecular diagnosis³¹ (see Fig. 36.2). These mutations lead to either decreased representation of *MCP* proteins on the cell surface or to impaired ability to regulate complement on the surface of endothelial cells.¹⁹ The risk of posttransplant recurrence is low, as donor endothelial cells would presumably have normal amounts/function of *MCP*. However, rare cases of recurrence have been reported, possibly explained by repopulation of the allograft vasculature with recipient endothelial cells, a phenomenon known as endothelial microchimerism.⁴³

Complement factor I-associated aHUS. *CFI* is the third major negative regulator of the complement system¹ (see Fig. 36.1). It was therefore not surprising when two teams reported that *CFI* was frequently mutated in patients with aHUS.^{44,45} Incomplete penetrance and variable expressivity are the norms in *CFI*-associated aHUS. Data suggest that ~10% to 15% of aHUS patients have a heterozygous *CFI* mutation³¹ (see Fig. 36.2). As with other negative complement regulators, *CFI* mutation leads to low *CFI* protein levels or reduced enzymatic activity, thereby causing higher activation of the alternative pathway.⁴⁶

C3-associated aHUS. Complement component 3 (*C3*) is central to the alternative complement cascade as it is the substrate required to generate the bioactive fragments C3b, AP C3 convertase and C3a¹ (see Fig. 36.1). Pathogenic heterozygous *C3* mutations, which were first reported in 2008, result in a gain-of-function phenotype by preventing the degradation of C3b (e.g., via reduced binding to *CFH* or *MCP*).⁴⁷ In some rare cases, the gain-of-function phenotype is due to enhanced C3 binding to *CFB*.⁴⁸ About 10–15% of aHUS patients harbor *C3* mutations³¹ (see Fig. 36.2).

Complement factor B-associated aHUS. Cleavage of factor B (*FB*) by factor D (*FD*) is necessary to produce the C3 convertase, C3bBb¹ (see Fig. 36.1). Gain-of-function *CFB* mutations cause either enhanced formation or delay the decay of C3 convertases.⁴⁹ It is now clear that this is a rare form of aHUS, accounting for ~2% to 4% of all confirmed cases of aHUS³¹ (see Fig. 36.2). *FB*-associated aHUS cases are generally fully penetrant,⁵⁰ but relatives consistent with incomplete penetrance have also been described.⁴⁹

Genetic forms of aHUS and TMA unrelated to complement

Mutations in genes that are not a component of the complement system, in particular the alternative pathway, have also been reported to cause aHUS. These include cobalamin C (*MMACHC*),³⁴ thrombomodulin (*THBD*),⁵¹ *DGKE*,^{52,53} and plasminogen (*PLG*).³⁰ There is currently no clear evidence that therapies targeting the complement system are effective in these conditions unless patients also carry a mutation in one of the complement genes. It is also unclear if these genes play a role through their interaction

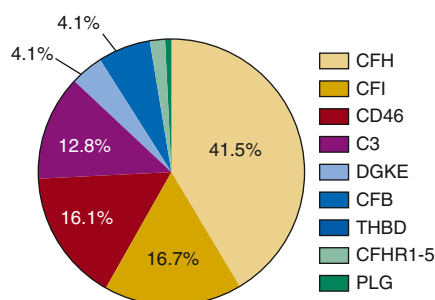


Fig. 36.2 Yield of genetic testing for a large cohort of patients with clinical features of atypical hemolytic uremic syndrome (aHUS). The data presented are from Osborne et al.,²² which collated genetic testing results from 3108 patients from 6 centers. The top panel illustrates the percentage of positive tests for given aHUS genes when compared with all positive tests. This data are also displayed graphically in the pie chart in the bottom panel.

with complement or whether a coagulation protein-specific mechanism is involved.

Thrombomodulin-associated aHUS. Although previously mostly known as a cofactor for thrombin, preventing blood coagulation⁵⁴ thrombomodulin (*THBD*; CD141) has been demonstrated to regulate CFI-dependent degradation of C3b. These studies were triggered by the discovery of an association between heterozygous *THBD* mutations and aHUS more than 10 years ago.⁵¹ *THBD* mutations now account for ~1% to 2% of aHUS patients³¹ (see Fig. 36.2). There is also some evidence linking *THBD* mutation with increased risks of venous thrombosis or myocardial infarction.⁵⁵ The pathophysiology of this condition is not well understood. Of note, the interpretation of the significance of the pathogenicity of *THBD* mutations in aHUS is currently debated.⁵⁶

Cobalamin C deficiency-associated aHUS. With only ~40 cases reported since 1992, cobalamin C (cblC) deficiency-associated HUS is a rare form of TMA.⁵⁷ This autosomal recessive condition is caused by mutations in the gene encoding for methylmalonic aciduria and homocystinuria type C protein (*MMACHC*).^{34,58} Patients with this condition typically present as neonates with failure to thrive, neurologic symptoms, and features of TMA. High mortality has been reported for infantile cblC-associated HUS.^{57,59} A subset of patients present as teens,⁵⁹ and adult onset is unusual.⁶⁰ The diagnostic hallmarks are hyperhomocystinemia, low blood methionine, and methylmalonic aciduria.⁶¹ Patients are usually treated with a combination of hydroxocobalamin, folinic acid, and betaine, though the impact of this cocktail on neurologic and vascular pathology is variable.⁶² Though not completely understood, several mechanisms are proposed to explain the TMA lesions, including hyperhomocystinemia-induced endothelial damage or impairment of nitric oxide-dependent inhibition of platelet aggregation.^{59,61}

Diacylglycerol kinase epsilon-associated aHUS. First reported in 2013,⁴³ DGKE-associated aHUS is one of the most common forms of aHUS diagnosed in children younger than 2 years old. Patients with this condition experience TMA recurrences until 5 years of age (usually triggered by infection), have increasingly worse hypertension and proteinuria with age, and develop kidney failure between the ages of 10 and 25.^{53,63} The clinical features do not improve with C5 inhibition, and there are no documented cases of posttransplant aHUS recurrence. There are now more than 60 patients diagnosed with this ultrarare condition^{53,63}; patients with DGKE nephropathy account for ~4% to 5% of all aHUS patients³¹ (see Fig. 36.2). Patients with recessive pathogenic genotypes show almost complete deficiency of DGKE activity. About 5% to 10% of patients with clearly pathogenic DGKE genotypes present with an active glomerulopathy but without findings of TMA,⁶⁴ a phenomenon that remains unexplained. While the exact pathophysiology remains elusive, it appears that DGKE deficiency causes endothelial cell activation,⁶⁵ perhaps by interfering with prostaglandin-E2 production⁶⁶ or signaling via vascular endothelial factor receptor-2.⁶⁷

Plasminogen-associated aHUS. Since thrombosis is central to TMA, a group of patients with undiagnosed aHUS were

screened for mutations in all major genes implicated in the coagulation pathway using targeted exome sequencing.³⁰ The most notable finding was that 3/36 patients had heterozygous plasminogen (*PLG*) variants, all of which were known to cause *PLG* deficiency in the homozygous state. One study revealed that ~1% of patients with aHUS have a *PLG* variant³¹ (see Fig. 36.2). In principle, *PLG* deficiency could reduce dissolution of thrombi, leading to a prothrombotic state.⁶⁸ The same question was the subject of intense investigations 20 years ago for patients with venous thromboembolism where the overwhelming conclusion was that heterozygous *PLG* deficiency is a benign condition.⁶⁹ It remains to be established whether *PLG* variants are directly disease-causing in aHUS or act more as risk factors.

tRNA splicing endonuclease-associated aHUS. An intronic variant in the gene *TSEN2*, which results in abnormal splicing of the mRNA of this gene, has been found in six individuals from four consanguineous families. Those patients presented with aHUS, as well as microcephaly, multiple craniofacial malformations, radiologic abnormalities of the central nervous system, and cognitive retardation of variable severity and progressed quickly to kidney failure in childhood.⁷⁰ It is unclear how this mutation leads to the TMA phenotype.

Inverted formin 2-associated aHUS. In a U.K. cohort, aHUS was described in two families with mutations in *INF2*, a gene encoding for a formin protein involved in actin polymerization and depolymerization.⁷¹ Additionally, all individuals affected had common aHUS risk haplotypes. Therapy with complement blockade was unsuccessful.⁷¹ Mutations in *INF2* lead to Charcot-Marie-Tooth disease (present in 2 individuals) and cases with Charcot-Marie-Tooth and focal segmental glomerulosclerosis have been described.⁷²

Autoimmune forms of aHUS—CFH and CFI autoantibodies

The discovery that mutations in many complement genes caused aHUS led to the hypothesis that autoantibodies directed against the complement proteins could also trigger an autoimmune form of aHUS.^{73,74} In 2005, investigators provided the first demonstration that anti-CFH autoantibodies were present in a subset of patients with aHUS, resulting in an acquired functional CFH deficiency.⁷⁵ The antibodies bind to the C-terminus and interfere with the binding of CFH to the cell surface.⁷⁵⁻⁷⁷ Of interest, most patients with anti-CFH autoantibodies have homozygous deletions of *CFHR1* and *CFHR3*, suggesting that gene deletions may play a role in the development of autoantibodies.⁷⁸⁻⁸⁰ The functions of CFHR proteins and their role in autoantibody production is not well understood.⁸¹

In addition, in patients with aHUS, anti-CFI autoantibodies have been identified, which also cause activation of the complement system.⁸² Anti-CFH and anti-CFI autoantibodies typically account respectively for ~10% to 25% and ~2% of aHUS cases, except in India, where anti-CFH autoantibodies are the chief cause of aHUS and occur in ~55% of aHUS patients,⁸³ and anti-CFI autoantibodies are found in ~30% of patients.⁸⁴ The reason for this is not well understood. Interestingly, anti-C3 autoantibodies enhancing the activity

of the (alternative pathway) C3 convertase upon binding (so-called C3 nephritic factor, C3NeF) are found in patients with C3G but have not been reported as a cause of aHUS.⁸⁵ Patients with an autoimmune pathogenesis of aHUS often show a relapsing course.^{6,83,86} Higher titers of CFH antibodies have been associated with a high risk of relapse.⁸³ Treatment guidelines recommend either prompt initiation of therapy with plasma exchange (PLEX) and immunosuppressive medications but lately favor the use of eculizumab.^{23,87,88} There have been no studies that compare the efficacy of C5 inhibition with PLEX. Several case reports demonstrate the successful use of eculizumab either after failed PLEX or as initial treatment.^{89,90} Five to seven PLEX sessions achieve 80% reduction in antibody titers.⁹¹ Immunosuppressive therapy as induction therapy mainly includes a short course of steroids plus either mycophenolate mofetil (MMF), cyclophosphamide, or rituximab.^{6,86-88,92,93} No data favor one over the other, and the choice of immunosuppression will depend on treatment availability and how comfortable clinicians feel with either treatment option. CFH antibody titers should guide treatment and once undetectable for at least 6 months, KDIGO suggests that treatment could be discontinued.⁸⁸ Data from case reports and a large series of patients from India indicate that long-term mycophenolate mofetil helps to reduce overall relapse risk and improve kidney outcome.^{6,92,93} Similarly, long-term MMF therapy allows for cessation of C5-inhibitory therapy in those patients, important also given the cost of this therapy.⁹⁴ Overall outcomes have been good for patients with CFH antibody-associated aHUS if treated with PLEX and long-term immunosuppression or C5 inhibition. However, kidney failure has been reported in about 10% to 25%, and late initiation of treatment is associated with worse outcomes.^{6,83,89,92,93}

Secondary forms of aHUS and aHUS in the context of complement-amplifying conditions

As mentioned earlier, several conditions or other diseases can cause transient complement activation and/or endothelial injury leading to TMA. Diagnosis is made by the identification of a presumed trigger of endothelial cell activation or complement activation and the absence of disease-causing genetic variants that could explain the phenotype. [Table 36.1](#) shows a list of conditions that can result in aHUS. Complement targeting therapy should be considered on a case-by-case basis according to the consideration of the causal and triggering condition and possibly severity of presentation. Complement activation is usually transient, so only short-term complement-blocking therapy is required.⁴

TREATMENT OF aHUS

Atypical HUS as a disease entity has largely established itself as a disease of complement dysregulation. Its treatment should consider two components: 1. management of sequelae from AKI (HTN, proteinuria, oliguria, electrolyte disturbances and uremia), which is not covered in the scope of this chapter; and 2. treatment of the underlying causative process.

The current gold standard is therapy with C5 inhibitors^{23,88}; however, in low-income countries plasma exchange or plasma infusions are often the only available or accessible treatment options.⁸⁷ Plasma exchange with immunosuppression and C5

inhibition are equally considered treatment options for CFH-autoantibody HUS, which has been lately favored by experts in the field.^{23,88}

Terminal complement blockade

Eculizumab is a recombinant humanized monoclonal antibody that binds C5 to prevent its cleavage and thereby formation of both C5a and C5b-9.⁹⁵ Subsequently, it reduces anaphylatoxin-induced inflammation and limits the prothrombotic consequences of C5b-9 activation. In clinical trials eculizumab therapy has resulted in substantial and sustained improvements in kidney function even in patient subgroups with long-standing, advanced kidney damage requiring kidney replacement therapy.^{95,96} Earlier administration (shorter interval between the current clinical manifestation of aHUS and initiation of treatment) is associated with greater improvements in estimated glomerular filtration (eGFR), reversal of organ damage, and health-related quality of life.⁹⁵ Due to the lack of a single diagnostic biomarker to rapidly confirm complement-mediated aHUS, eculizumab therapy should be commenced in suspected aHUS and discontinued if an alternative diagnosis is identified. Eculizumab is approved for the treatment of aHUS in the United States, Canada, and Europe with centers utilizing weight-based dosing adapted from trial data and manufacturer specifications⁹⁵ ([Table 36.2](#)). Dosing intervals are based on drug pharmacokinetics and should be kept within 48 hours of recommended time points. Monitoring with complement functional testing (CH50 or AP50) is advised in general, but especially if patients show an incomplete response.

Ravulizumab is a long-acting form of anti-C5 therapy derived from eculizumab with the change of 4 amino-acids designed to increase drug half-life and thus dosing intervals.⁹⁷ Ravulizumab, typically dosed every 8 weeks, is effective in aHUS treatment for pediatric and adult patients both in treatment naïve and on chronic eculizumab therapy subgroups.⁹⁷⁻⁹⁹ Increased dosing intervals decreases the treatment burden on patients and medical day-stay units. However, as patents on eculizumab expire, the emergence

Table 36.2 Eculizumab Dosing Regimen (Induction and Maintenance)

Body Weight	Induction	Maintenance
40 kg and above	900 mg weekly for 4 weeks	1200 mg at week 5; then 1200 mg every 2 weeks
30 to <40 kg	600 mg weekly for 2 weeks	900 mg at week 3; then 900 mg every 2 weeks
20 to <30 kg	600 mg weekly for 2 weeks	600 mg at week 3; then 600 mg every 2 weeks
10 to <20 kg	600 mg week 1	one dose 300 mg at week 2; then 300 mg every 2 weeks
5 to <10 kg	300 mg week 1	one dose 300 mg at week 2; then 300 mg every 3 weeks

of cheaper biosimilars may challenge the opportunity cost of ravulizumab in the future.¹⁰⁰

Anti-C5 therapy is generally well tolerated; however, patients should be warned of the risks of infection with encapsulated organisms including all meningococcal strains, *S. pneumoniae*, and *Hemophilus influenza*. It is important that patients are vaccinated against the stated pathogens as quickly as possible if not already completed. If vaccination is not possible or delayed (>2 weeks), then prophylactic antibiotics (amoxicillin, penicillin or erythromycin if allergic) should be prescribed. Continued prophylactic antibiotics should also be considered in at-risk populations (<5 years old, additionally immunosuppressed individuals) as the efficacy of meningococcal vaccinations are unclear in the setting of complement blockade. Patients receiving anti-C5 treatment should be advised of the risk of meningococcal infection, albeit low, instructed on early identification of warning signs and symptoms, and provided with an information card to carry at all times, allowing for prompt recognition of their disease and treatment upon hospital admittance.²³

Complexity arises when deciding on the duration of therapy. Terminal complement blockade is not a disease-modifying approach as it does not eliminate the root cause of complement dysregulation in aHUS. Aside from anti-CFH antibody-driven disease and complement-independent forms such as DGKE-HUS, patients with complement-mediated forms (with and without genetic mutation) are always at risk of recurrence upon cessation of complement blockade. The risk of infection by encapsulated organisms must be balanced with the risk of disease relapse with subsequent irreversible kidney damage. Therefore if, how, and when to discontinue complement blockade is a difficult issue without a single clear answer. The type of complement mutation should be considered, which can confer some prognostication. Prospective trial data evaluating eculizumab withdrawal showed relapse rates of 23%, identifying the three main risk factors for relapse to be female gender, presence of rare complement gene variant, and increased soluble C5b-9 plasma levels at the time of discontinuation.¹⁰¹ The risk of relapse is even higher following kidney transplantation.^{102,103} Regardless of optimal timing, discontinuation of eculizumab therapy should be accompanied by a robust follow-up plan that includes close monitoring of thrombotic microangiopathy biomarkers (hemoglobin, platelet count, LDH, haptoglobin, blood film) and proteinuria. Eculizumab therapy should be promptly reinstated with the earliest signs of recurrence.

Special consideration is required for patients with antibody-driven aHUS. In the setting of anti-CFH antibody-driven aHUS, plasma exchange with concurrent immunosuppressive medications (cyclophosphamide, rituximab, or mycophenolate mofetil) represents an established mode of treatment. However, complement blockade with eculizumab can be effective at disease presentation as rescue therapy to induce remission,^{63,90,104,105} in addition to antibody titer reduction strategies such as long-term MMF.^{83,92} Anti-C5 therapy can likely be safely discontinued when patients have a negative/low CFH-FH-antibody titers and are continued on MMF. However,

investigations are ongoing regarding a “safe” titer level for treatment discontinuation and the duration of complement blockade required.

Plasma Exchange and Plasma Infusion

Before the advent of terminal complement blockade therapy, plasma therapy (infusion or exchange) was the mainstay of treatment with the rationale of replacing nonfunctioning complement proteins or removing FH-CFH-autoantibodies and hyperfunctional components of complement. Treatment efficacy was initially based on reports of successes in familial-related aHUS with plasma therapy, resulting in complete or partial remission in 78% of aHUS episodes in children and 53% in adults. However, follow-up investigation revealed that 48% of children and 67% of adults either died or reached kidney failure at 3-year follow-up.¹⁵ Outcomes were not correlated with dose administration, as both children and adults had poor kidney outcomes after the first episode of aHUS regardless of receiving high-dose plasma therapy (>5 plasma exchanges or plasma infusions >10 mL/kg per day for >5 days) or not.¹⁴ Consensus guidelines published in 2009 by The European Pediatric Study Group for HUS advocate for early and intensive plasma exchange during the first month of diagnosis.¹⁰⁶ In the pediatric population, especially, plasma therapy is fraught with complications including infection, thrombosis, and hemorrhage.¹⁰⁷ Despite showing efficacy at improving hematologic parameters, it has not shown to be effective at facilitating long-term kidney health. Despite this, due to the lack of available alternative therapeutic options in certain resource poor health systems, plasma therapy continues to be the mainstay of treatment.⁸⁷

As expected, complement-independent forms of aHUS such as DGKE-HUS have not shown to be responsive to plasma therapy or complement blockade.⁵²

PROGNOSIS

Before the introduction of anti-C5 therapy, the prognosis for aHUS patients was poor. The 1-year mortality was higher in children (6.4%) compared adults (0.8%), but conversely, progression to kidney failure after the first aHUS episode was more frequent in adults at 46% (vs. 16% in children).¹⁴ In totality, 56% of adults progressed to kidney failure by 1 year and 48% of children by 5 years. Prognosis varies by genotypic subgroups with MCP mutations carrying the best prognosis if the first episode occurrence was in childhood despite subsequent frequent relapses (25% kidney failure with median follow-up of 17.8 years). Mutations in *CFH*, *CFI*, and *C3* carry the worst prognosis¹⁵ with up to 60% of individuals with *CFI* or *C3* mutations and 77% with *CFH* mutations developing either kidney failure or death at 5 years.¹⁴ The risk of relapse after the first episode is ~40% with most relapses in adults occurring in the first year. The risk of relapses after the first year decreases to approximately 25% in all patients except children with MCP-associated HUS.¹⁴ The advent of anti-C5 therapy has drastically improved these statistics with only 10% to 15% of patients progressing to kidney failure at 1 year regardless of genetic mutation.¹⁰⁸ DGKE-mediated aHUS follows a relapsing/remitting course commonly progressing to CKD and kidney failure within 25 years of diagnosis.^{53,63}

TRANSPLANTATION IN aHUS PATIENTS

Before the era of terminal complement blockade, the overall outcome of kidney transplantation in adults with aHUS was poor with 7% of deaths and 50% of graft failure at 5 years post transplant. This is heavily impacted by the recurrence of disease usually in the first year post transplant leading to 5-year graft survival rate of 30% as opposed to 68% in patients without recurrence.¹⁰⁹ Most patients (~80%) who had lost a prior graft to recurrence had recurrence after retransplantation. Complement mutations heavily dictated the risk of recurrence. CFH-related aHUS cases were most likely to recur in almost 90% of patients followed by C3, CFB, and CFI genotypes with a 50% recurrence risk. MCP carries a favorable prognosis with only a 10% recurrence rate, as does anti-FH antibody-associated HUS dependent on antibody titer suppression. Mutations in MCP carry a low recurrence rate as it is a membrane-tethered protein as opposed to fluid phase complement proteins; thus renal allografts will protect against recurrence. Patients with CFH antibody aHUS seem to have a favorable outcome post transplantation if transplant was performed with negative or low CFH antibody titer,⁸³ probably due to the use of immunosuppressive medication post transplant. Posttransplant recurrence has not been observed in 6 patients with DGKE-HUS.^{53,63} One patient with *INF2* mutation presented with a TMA post kidney transplant.⁷¹

The high risk of recurrence in most patients with aHUS has led to the common practice of using anti-complement therapy prophylactically. Prophylactic anti-C5 therapy is recommended, especially for patients with a high risk for recurrence due to a high-risk mutation (CFH, C3, *FB*) or previous graft loss, but also for patients with moderate risk (no mutation, *CFI* mutation). For patients with low risk, such as those with an *MCP* mutation or a likely non-complement-mediated form of aHUS (e.g., drug-induced aHUS, infection-induced aHUS), an individual decision needs to be made depending on local professional experience, sometimes complemented with consultations with international colleagues, and in conjunction with the patient or family.⁸⁸ If anti-C5 therapy is not available, then peritransplant plasma exchange is recommended, although this has been less successful compared with C5 inhibition. Plasma therapy post transplantation has also failed to prevent graft loss in recurrence.⁴² With the use of prophylactic eculizumab, patients with moderate and high risk have decreased rates of recurrence and significantly increased rates of graft survival.¹⁰² For patients with CFH antibody HUS, it is recommended to lower CFH antibody levels before transplantation to either a negative or very low titer, at which juncture it is probably safe to perform kidney transplant without prophylactic anti-C5 therapy. Alternatively, perioperative PLEX can be performed to lower antibody titers.^{23,87}

Living-related transplantation has been considered a contraindication in aHUS¹¹⁰ because of the high risk of recurrence and the potential risk of aHUS in the donor triggered by the surgical procedure.¹¹¹ Data from the Netherlands, however, suggest that living donation can be safely performed in aHUS.¹¹² With the current state of knowledge, we recommend that living donation can be considered after genetic screening of the donor.⁸⁸ Living-related transplants have been performed in patients with

CFH antibody aHUS without recurrence in the recipient and with no issues for the donor within a 10-year observation window.¹¹³

FUTURE THERAPIES FOR aHUS

Multiple new therapies targeting various components of the complement pathway are in clinical trials (phase II or III).^{114,115} Terminal pathway inhibitors include crovalimab, a monoclonal antibody against C5, which is currently being evaluated in two global, single-arm, phase III studies in adult/adolescent and pediatric patients (COMMUTE-a and COMMUTE-p) with aHUS.¹¹⁶ Crovalimab has a half-life of 30 days and was engineered to be administered in small-volume subcutaneous injections every 4 weeks (2 weeks for patients weighing <20 kg). Data also suggests an involvement of the complement lectin pathway in TMA. Narsoplimab, a drug that suppresses the lectin pathway effector enzyme, mannose-binding lectin-associated serine protease (MASP2), is currently being tested in phase III studies.^{117,118} Avacopan (CCX168) is an orally administered C5a receptor (C5aR) inhibitor currently in phase II studies. Engineered to prevent C5a/C5aR1 axis activation while preserving C5b-9 formation, it is designed to counteract prothrombotic effects without diminishing the terminal complement effects against infection.^{119,120}

THE SPECTRUM OF THROMBOTIC MICROANGIOPATHY

Several conditions (see Table 36.1) have been associated with the clinical picture of TMA.⁵ Historically they were all termed secondary HUS, but research has shown a more prominent role for the complement system in some but not others. As this distinction has treatment implications, the best nomenclature is to term the condition TMA associated with a given “condition,” such as TMA associated with stem cell transplantation. We think of TMA as a spectrum, with aHUS being a primary complement-driven disease at one end and TMAs associated with conditions without complement involvement, such as drug-induced TMA at the other.

There are two ways in which TMA can develop in the setting of a coexisting disease and/or in association with medications:

- The patient has an underlying complement disorder/genetic susceptibility, and the disease/drug triggered the onset or recurrence of the disease.⁵
- The patient has no underlying complement disorder/genetic susceptibility but the disease/drug causes endothelial cell activation, resulting in endothelial cell injury and the clinical picture of TMA. Complement activation might also occur in this setting secondary to endothelial cell activation. Short-term complement blockade might be helpful in these cases.

Some conditions like pregnancy are more commonly a trigger event, and the likelihood of finding an underlying complement abnormality is high.²⁴ Other conditions such as malignant hypertension and drugs are more likely to result in endothelial cell injury and secondary complement activation.⁵ Some other conditions are less well defined. We currently do not have reliable routinely available

biomarkers that can help to distinguish genetic TMA from TMA associated with coexisting diseases. Low C3 can be observed in some patients with genetically driven aHUS but is not reliable.¹⁵ Similarly, C3d and sC5b-9 might help determine systemic complement activation, but half of the patients with atypical HUS have normal levels.¹²¹

In general, clinical presentation includes microangiopathic hemolytic anemia, thrombocytopenia, and kidney disease, which can manifest as elevated creatinine, proteinuria/nephrotic syndrome or nephritis, or hypertension. Specific algorithms have been published for pregnancy-associated TMA¹²² and TMA post bone marrow/hematopoietic stem cell transplantation.¹²³

THROMBOTIC MICROANGIOPATHY ASSOCIATED WITH PREGNANCY

Pregnancy has long been recognized as a condition that can trigger a TMA episode. It is not surprising that complement defects were identified in 80% of women developing TMA during pregnancy and especially in the early weeks after delivery.¹²⁴ TTP, due to increased release of von Willebrand factor during pregnancy, is an important differential diagnosis, especially if TMA occurs in the second or third trimester.²⁴ Pregnancy complications including preeclampsia and HELLP syndrome are common in pregnancy-associated thrombotic microangiopathy (TMA), particularly in cases of pregnancy-triggered atypical hemolytic uremic syndrome (p-aHUS) as characterized by Fakhouri et al.¹²⁵ A consensus paper suggests the following criteria: 1. platelets count $<100 \times 10^9/L$, 2. hemoglobin <0 g/dL, 3. LDH >1.5 upper limit of normal, 4. undetectable serum haptoglobin, 5. negative direct erythrocyte antiglobulin test, and 6. schistocytes on the blood smear of features of TMA on kidney biopsy.¹²² All patients should receive ADAMTS13 testing and complement genetics. Once ADAMTS13 testing is back and TTP has been ruled out, patients should be commenced on anticomplement therapy, which has been given to several patients during pregnancy and is deemed safe for the fetus. A minimum treatment duration of 6 months was suggested. Special considerations need to be given to aHUS patients who want to get pregnant. In patients with a previous TMA episode during pregnancy/postpartum, prophylactic anticomplement therapy can be considered.¹²⁶ These women must be closely monitored for at least 3 to 4 months postpartum. In the case of a relapse, anticomplement therapy should be initiated promptly. Pregnancy-associated TMA has a higher miscarriage rate, and one third of the babies showed a low birth weight.^{127,128} For further discussion on pregnancy, see [Chapter 58](#).

THROMBOTIC MICROANGIOPATHY ASSOCIATED WITH SOLID ORGAN TRANSPLANTATION

TMA post kidney transplantation can occur as a recurrence of aHUS or de novo and has been described in 1% to 14% of recipients. Following kidney transplantation, de novo TMA can be the result of a complement disorder, where the transplantation served as the trigger for aHUS, but it can also be secondary to several peritransplant and posttransplant phenomena such as endothelial cell injury, infections, alloimmune reactions, ischemia-reperfusion injury, and

drug effects.¹²⁹ TMA has also been described in about 2% to 4% of recipients of other solid organ transplants, with similar risk factors as described earlier.¹³⁰ Calcineurin inhibitors are frequently used as part of the immunosuppressive regimen and have been implicated in the pathophysiology of de novo TMA following transplantation. CNI results in renal arteriolar vasoconstriction, platelet aggregation, and endothelial injury.^{129,131,132} The use of mTOR inhibitors, sirolimus and everolimus, have also been proposed to cause de novo TMA, as they decrease vascular endothelial growth factor (VEGF) expression.¹³³ The highest rates of TMA have been reported in patients who received both a CNI and an mTOR inhibitor post transplant.¹³⁴ About 30% of patients with de novo TMA post kidney transplant have genetic complement mutations. The role of complement is less defined in nonkidney solid organ transplantation. There are no specific diagnostic criteria. TMA should be considered in every patient with a rising creatinine. Not all patients with TMA post kidney transplant have thrombocytopenia and/or anemia and even if they do, it might be preexisting and have a different cause. A kidney biopsy is recommended¹³⁵ for diagnosis but also to rule out concomitant antibody-mediated rejection (ABMR) or other glomerular diseases such as SLE. TMA associated with ABMR has been associated with worse outcomes.¹³⁶ Treatment of recurrent aHUS post transplant includes the start of anticomplement therapy. New-onset TMA often requires a multilayered diagnostic and management approach, depending on what the treating physician considers the main trigger for the TMA. Any coexisting conditions that cause endothelial cell activation should be addressed first: treatment of underlying infection, treatment of arterial hypertension, and treatment of antibody-mediated rejection. If none of these factors are apparent, lowering or discontinuation or switch of immunosuppressive agent can be considered while gauging the risk for rejection. As 30% of patients with de novo TMA following kidney transplantation harbor complement mutations, initiation of anti-complement therapy is a reasonable approach. Such therapy should be promptly started in patients with 1. systemic or biopsy-proven TMA and impaired kidney function, 2. patients with family history of TMA, 3. unknown cause of primary diagnosis of kidney failure (could have been aHUS in the first place), and 4. patients in whom switching immunosuppression or discontinuation is not feasible. In centers without access to complement targeted therapy, PLEX should be considered as an option.

THROMBOTIC MICROANGIOPATHY ASSOCIATED WITH HEMATOPOIETIC STEM CELL TRANSPLANTATION

TMA post hematopoietic stem cell transplantation (HSCT) is a severe complication that has been reported in 10% to 30% of patients, with one third being severe in children. TMA usually develops within the first 1 to 2 months post transplant.¹²³ Regular screening for TMA including regular CBC, blood smear, LDH, kidney function, urine dipstick, and blood pressure measurements is advised. Patients often present with proteinuria and hypertension, and extrarenal symptoms are also common.

High-risk TMA according to the Cincinnati group¹²³ is diagnosed if one of the following three conditions is present:

- A. TMA confirmed on kidney biopsy OR
- B. Clinical diagnosis of TMA (5 out of 7 markers must be present and must include 6 and 7):
 1. LDH above normal for age
 2. Schistocytes on blood smear
 3. De novo thrombocytopenia or increased transfusion requirements
 4. De novo anemia or increased transfusion requirements
 5. Hypertension >99% for age (<18 years) or >140/90 mm Hg (18 years of age)
 6. Proteinuria 30 mg/dL measured twice; random urine protein/creatinine ratio >2 mg/mg
 7. Terminal complement activation (elevated plasma sC5b-9 above normal)
- C. Multiorgan dysfunction syndrome (at least one organ) and the presence of five of seven criteria with at least one high-risk marker present (either numbers 6 or 7 from criteria B).

A kidney biopsy is helpful in making the diagnosis but often not feasible. Anti-C5 therapy is considered first-line therapy for patients with high-risk TMA following HSCT. Treatment monitoring using CH50 or eculizumab levels, if available, should be conducted regularly as these patients often need more frequent dosing intervals (sometimes as often as every other day). Response rates are reported as high as 60% to 70% with an overall 1-year survival rate of 50% to 66% compared with 17% without treatment.^{123,137} Once the patient has stabilized, eculizumab therapy can be discontinued. Treatment of mild disease includes management of hypertension, decrease of immunosuppressive therapy, and treatment of infections. Although patients show much improved long-term survival, chronic kidney disease is still common.¹²³

THROMBOTIC MICROANGIOPATHY ASSOCIATED WITH AUTOIMMUNE DISEASES

TMA—both HUS and TTP—can occur in the setting of other autoimmune diseases, such as SLE, antiphospholipid syndrome, and antineutrophil cytoplasmic antibody vasculitis.¹³⁸ TMA in SLE has been reported to occur in up to 9% of cases, and TMA lesions on biopsies are reported even more frequently (25%).^{139,140} Making the diagnosis of TMA is often difficult as clinical symptoms overlap significantly. Biopsy is usually helpful. Complement defects are rare, but complement is already an established part of autoimmune pathophysiology and therefore it is not surprising that overactivation can lead to endothelial cell injury and TMA.^{141,142} Treatment is mainly guided by the underlying disease, but in refractory cases, PLEX and complement inhibition have been successfully attempted. Further discussion on scleroderma can be found in [Chapter 31](#).

THROMBOTIC MICROANGIOPATHY ASSOCIATED WITH DRUG EXPOSURE

TMA from drug exposure is either secondary to a toxin-mediated effect on the endothelium or a dose-related immunologic reaction. There are no markers that distinguish a drug-mediated TMA from other causes. The most commonly reported agents include chemotherapeutic agents (gemcitabine, mitomycin, cisplatin); calcineurin

inhibitors; tyrosine kinase inhibitors; and vascular endothelial growth factor inhibitors.¹⁴³ Treatment of TMA is guided by withdrawal of the potentially offending drug. Complement activation is rarely seen, except in CNI-induced TMA.

THROMBOTIC MICROANGIOPATHY ASSOCIATED WITH ADENO-ASSOCIATED VIRUS ADMINISTRATION FOR GENE THERAPY

TMA following systemic adeno-associated virus (AAV) administration has been described, including the death of a 6-month-old with spinal muscle atrophy harboring a factor I variant of unknown significance.¹⁴⁴ AAV particles activate the classical complement pathway (triggered by anti-capsid IgM and IgG antibodies) but also directly interact with C3 and activate the alternative pathway. TMA induced by AAV is a serious and potentially life-threatening complication. Patients receiving gene therapy need to be closely monitored.¹⁴⁵ Whether treatment with anticomplement agents is of benefit is unknown.

THROMBOTIC MICROANGIOPATHY ASSOCIATED WITH INFECTIONS

Shigatoxin is the most common reason for HUS/TMA in childhood (see [Chapter 71](#)). TMA has also been reported in the setting of bacterial, viral, and fungal infections (see [Table 36.1](#)). Pneumococcal HUS is the other common form of postinfectious TMA in children.¹⁴⁶ Importantly, patients with pneumococcal TMA have a positive Coombs test due to Thomsen-Friedreich antigen, which gets exposed after cleavage of *S. pneumoniae*-produced neuraminidase.¹³ Similarly, H1N1¹⁴⁷ and COVID-19¹⁴⁸ have been reported to cause TMA. Severe infections such as sepsis or endocarditis have been associated with TMA, probably due to endothelial injury.¹⁴⁹ Treatment of the infection and close monitoring of TMA parameters is recommended. In some severe refractory cases, short-term anticomplement therapy might be considered. Complement- and infection-associated glomerulonephritis is discussed further in [Chapter 34](#).

THROMBOTIC MICROANGIOPATHY ASSOCIATED WITH MALIGNANT HYPERTENSION

The link between malignant hypertension or hypertensive emergency and TMA is the subject of an ongoing debate. It remains unclear whether the TMA complicating malignant hypertension is due to an underlying complement defect or secondary to severe hypertension and associated endothelial cell injury. Hypertensive emergencies complicate an aHUS diagnosis (with 50% having genetic complement defects) in 50% of adult patients.²⁴ On the other hand, patients with malignant hypertension have only a small risk (3%) of developing TMA. Indicators for hypertension being more likely essential and not related to a TMA are male, >45 years, previous history of hypertension, no clots on kidney biopsy, no need for kidney replacement therapy, and quick recovery (<72 hours) of the TMA with strict blood pressure control.²⁴ A positive family history of aHUS or recurrent episodes of hematological features of TMA, however, could speak for aHUS and needs further investigation. Some patients with malignant hypertension and TMA do not have hematologic signs of TMA, but TMA lesions may be present on biopsy.¹⁵⁰ In pediatrics, TMA due

to malignant hypertension is rare and a child presenting with TMA and a hypertensive emergency is more likely to have aHUS and needs a complement workup. The first steps include treatment of the hypertensive emergency and close monitoring of the TMA. Anticomplement therapy can be considered, especially if there is no rapid recovery after initiation of antihypertensive therapy.

C3 GLOMERULOPATHY AND RELATED CONDITIONS

INTRODUCTION

C3 glomerulopathy (C3G) and immune-complex MPGN (IC-MPGN) are characterized by proteinuria, hypertension, and impaired kidney function, and patients are at significant risk of progression to kidney failure in roughly 20% in children and 50% in adults within 10 to 15 years from onset.¹⁵¹ Kidney biopsy shows membranoproliferative injury pattern with C3 and/or immunoglobulin deposits.¹⁵¹ Complement plays a significant role in disease pathogenesis for both. An important differential diagnosis is acute poststreptococcal glomerulonephritis (APSGN), a classical immune complex-mediated nephritis following a streptococcal skin or throat infection¹⁵² (discussed further in [Chapter 34](#)). Complement C3 is low secondary to binding to immune complexes. All three diseases are discussed as follows.

C3 GLOMERULOPATHY AND IC-MPGN

CLASSIFICATION

The current classification is based on biopsy findings, determined by the presence or absence of immunoglobulins and complement components on renal biopsy: 1. C3 glomerulopathy (C3G) has evidence of C3 deposits alone or C3 dominant deposits, and 2. immune-complex MPGN (IC-MPGN) has C3 deposits with dominant or codominant immunoglobulins/immune complex deposits.^{153,154} Of note, membranoproliferative glomerulonephritis without immune reaction on immunofluorescence (IgG and C3 negative) can occur as a result of endothelial injury secondary to diseases such as HUS, chronic sickle cell anemia, connective tissue disease, and transplant glomerulopathy.¹⁵⁵ C3G is further divided depending on electron microscopy findings into C3 glomerulonephritis (C3GN) and dense deposit disease (DDD), the latter showing characteristic ribbon-like deposits within the glomerular basement membrane.¹⁵⁵

At the time when the new classification consensus was published in 2012,¹⁵⁴ it was thought that this classification would split cases of MPGN into complement-mediated (C3G) and secondary (IC-MPGN) forms ([Table 36.3](#)). However, this hasn't held true as autoantibodies and complement mutations are found in C3G and IC-MPGN patients. Similarly, secondary causes can lead to both C3G and IC-MPGN. This indicates that MPGN is a heterogeneous spectrum rather than a distinct disease and most likely a continuum of an IC-MPGN to C3-GN spectrum.^{156,157} Understanding the underlying pathophysiology is important as therapies must target either complement activation or the immune system. As the kidney biopsy cannot distinguish

between complement-mediated versus non-complement-mediated forms, screening for autoantibodies and genetic alterations is of utmost importance.

The primary focus in this chapter is on complement-mediated forms of MPGN, for both C3G and IC-MPGN cases.¹⁵³⁻¹⁵⁵ Secondary causes of IC-MPGN are discussed elsewhere in this book.

HISTOPATHOLOGY

Light microscopy: The four major morphologic patterns seen in DDD and C3GN include a mesangial proliferative pattern, a membranoproliferative pattern, a crescentic pattern, and an acute proliferative and exudative pattern

Table 36.3 Secondary Causes of IC-MPGN

Immunoglobulin mediated

Infectious diseases

- Hepatitis B, hepatitis C, hepatitis E, Epstein Barr virus, and human immunodeficiency virus
- Endocarditis, shunt nephritis, visceral abscess, and empyema
- Tuberculosis, leprosy
- Brucellosis, Q fever
- Malaria, schistosomiasis
- Onchocerciasis; filariasis, *Loa loa* infections

Systemic immune diseases

- Cryoglobulinemia
- Systemic lupus erythematosus
- IgA vasculitis
- Sjögren syndrome (secondary to cryoglobulinemia)
- Hypocomplementemic urticarial vasculitis
- Rheumatoid arthritis
- X-linked agammaglobulinemia
- Neoplasms/dysproteinemias
- Plasma cell dyscrasia
- Monoclonal gammopathy of unknown significance
- Fibrillary glomerulopathy
- Immunotactoid glomerulonephritis
- Proliferative glomerulonephritis with monoclonal immunoglobulin deposition
- Proliferative glomerulonephritis with isolated light chain deposition
- MPGN with masked monotypic immunoglobulin deposits
- Light-chain and/or heavy-chain deposition disease
- Leukemias and lymphomas (with or without cryoglobulinemia)
- Bone marrow transplant
- Carcinomas, Wilms tumor, and malignant melanoma
- Angiofollicular lymph node hyperplasia with or without TAFRO syndrome
- Sinus histiocytosis with massive lymphadenopathy

Other

- α -1-antitrypsin deficiency with severe liver disease
- Gluten-sensitive enteropathy (celiac disease)
- Immunoglobulin negative
- Thrombotic microangiopathy including drug-induced forms
- Sickle-cell disease
- Transplant glomerulopathy
- POEMS syndrome

(Figs. 36.3 and 36.4).¹⁵⁸⁻¹⁶⁰ In DDD, there is an eosinophilic and refractile appearance of the glomerular basement membrane.¹⁶¹

Immunofluorescence microscopy: C3 is dominant and positive in the glomerular basement membranes and mesangial regions. These deposits are seen along tubular basement membranes and the basement membrane of the Bowman capsule in DDD. It is important to note that immunoglobulins and C1q are not uncommonly seen in C3G in a pattern similar to the C3 deposition.¹⁵⁸⁻¹⁶⁰

Electron microscopy: The essential diagnostic feature of dense deposit disease is the presence of electron dense transformation of the glomerular basement

membranes,¹⁵⁸⁻¹⁶⁰ which looks as though they have been calligraphed. In many areas, the entire length of the GBM will be replaced, while elsewhere only segments may be involved. Usually, this transformation involves the full thickness of the loop but occasionally it only includes a part. Subepithelial humplike deposits with similar deep electron density are not uncommon.¹⁵⁸⁻¹⁶⁰ In the deep mesangium, this same electron-dense transformation assumes a spheroid appearance. In C3GN, there are light, amorphous mesangial, subendothelial, and subepithelial deposits seen on electron microscopy.¹⁶² In DDD, osmiophilic dense deposits are found in the lamina densa of the glomerular basement membrane.¹⁶¹

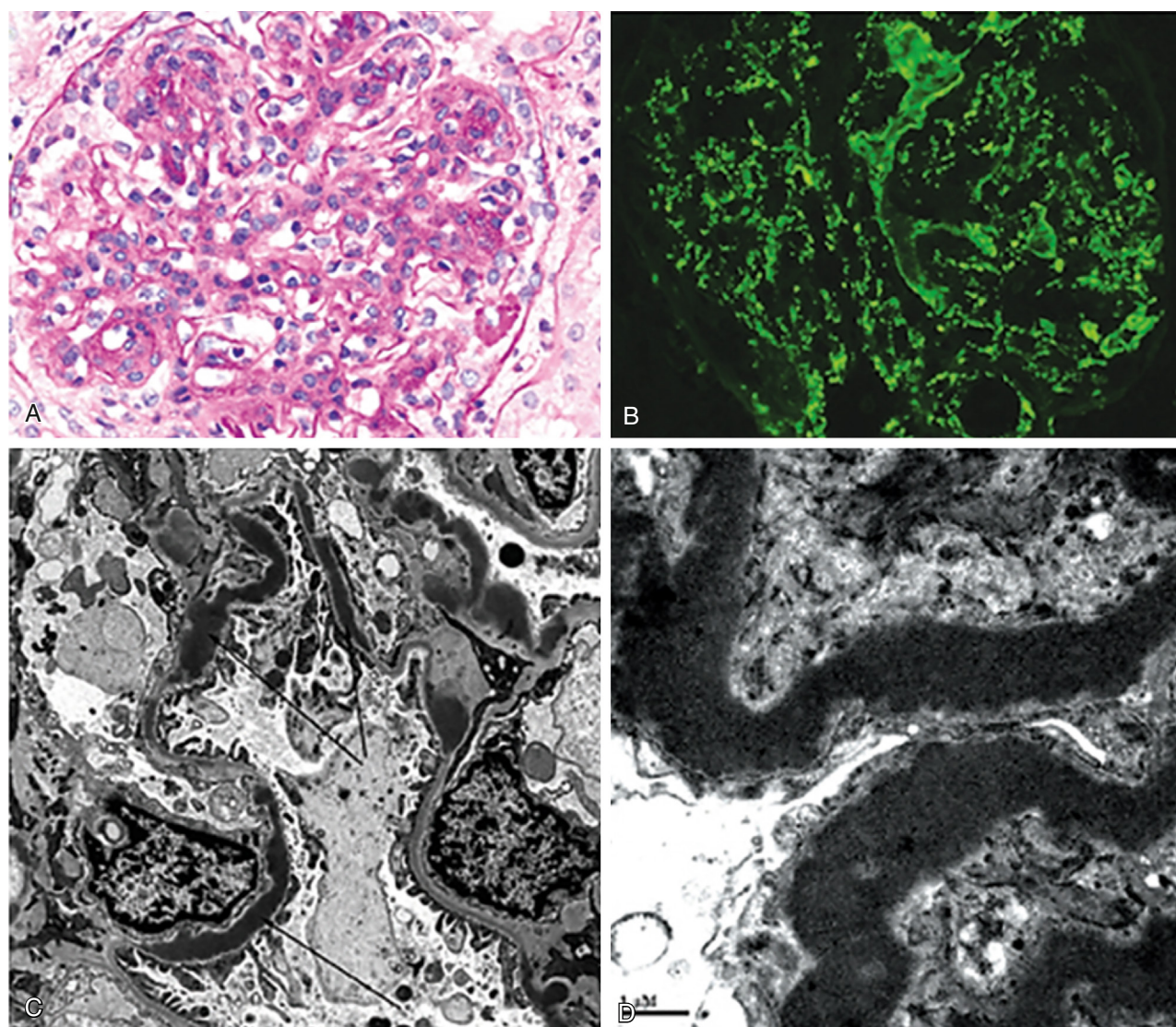


Fig. 36.3 Dense deposit disease. (A) Diffuse mesangial hypercellularity with segmental endocapillary hypercellularity and pink ribbon-like enhancement of glomerular basement membranes (periodic acid-Schiff stain, original magnification 600 $\times$); (B) C3 granular deposits diffusely in mesangial areas and segmentally along glomerular basement membranes (fluorescein-conjugated antihuman C3, original magnification 600 $\times$); (C) segmental electron dense transformation of glomerular basement membranes and mesangial areas (*arrows*, unstained grid, original magnification 4000 $\times$); and (D) diffuse electron dense transformation of glomerular basement membranes (unstained grid, original magnification 12,000 $\times$).

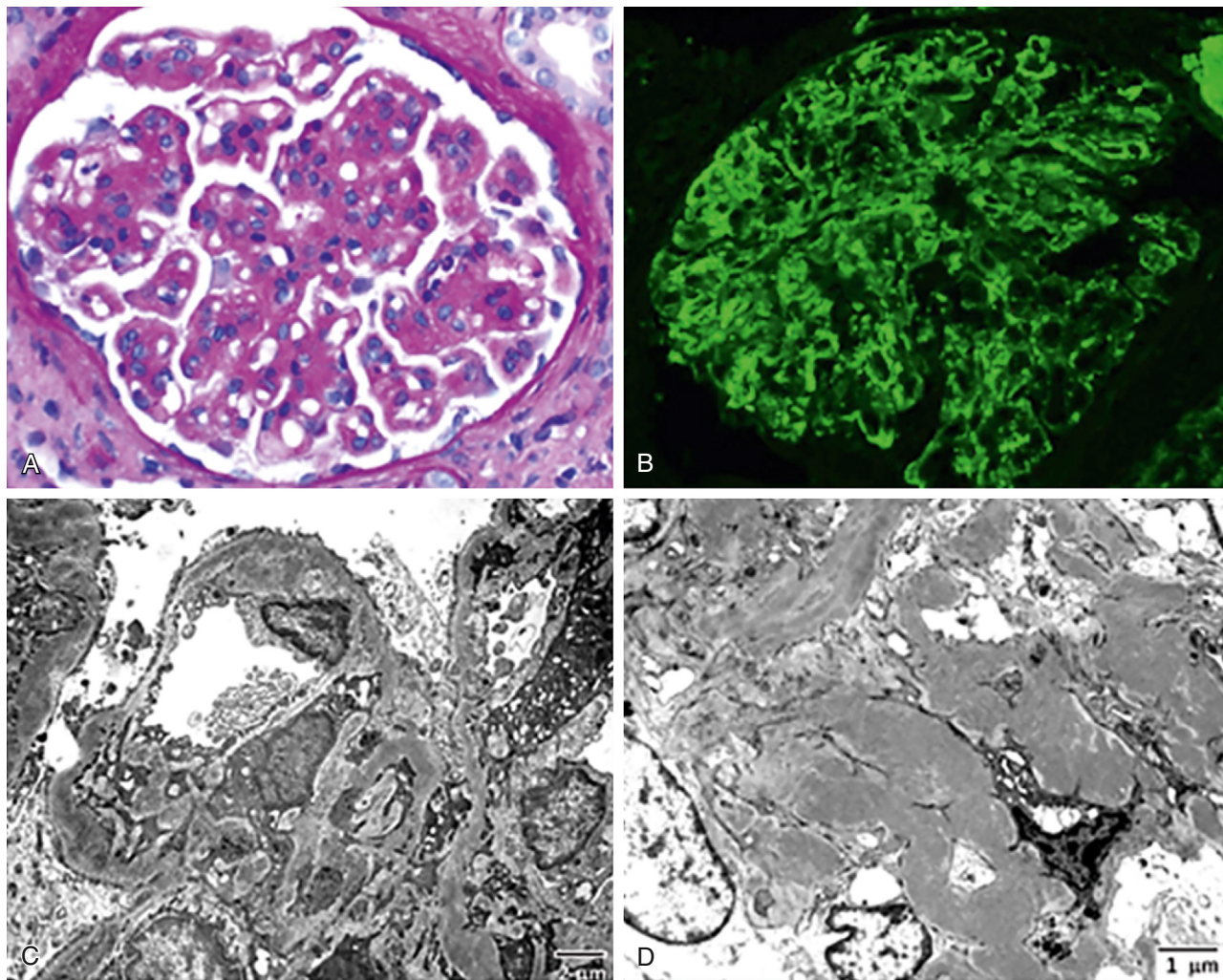


Fig. 36.4 C3 glomerulonephritis. (A) diffuse mesangial hypercellularity with endocapillary hypercellularity and thick glomerular basement membranes (periodic acid- Schiff stain, original magnification 600 $\times$); (B) C3 granular deposits diffusely in mesangial areas and along glomerular basement membranes (fluorescein-conjugated antihuman C3, original magnification 600 $\times$); (C) segmental charcoal gray transformation of glomerular basement membranes and mesangial areas (unstained grid, original magnification 4000 $\times$); and (D) diffuse charcoal gray transformation predominantly in mesangial areas but also within glomerular basement membranes (unstained grid, original magnification 8000 $\times$).

COMPLEMENT DYSREGULATION

There are three possible mechanisms for complement dysregulation in patients with C3G, which all ultimately lead to enhanced activation of C3:

- Autoantibodies, which stabilize and prolong the decay of the complement alternative pathway C3 convertase, resulting in its overactivation.
- Mutations or autoantibodies to CFH or CFHR proteins that result in the absence or loss of the function of CFH resulting in the loss of control of the complement alternative pathway C3 convertase and its enhanced activity.
- Mutations in *C3* or *CFB* resulting in an exceedingly stable complement alternative pathway C3 convertase with prolonged decay and enhanced function.
- Some autoantibodies prolong the half-life of the complement classical pathway C3 convertase.⁸⁵ Some nephritic factors stabilize the (alternative or classical pathway) C5 convertase (C5NeF) and result in terminal

pathway activation and elevated C5b-9 levels.^{163,164} The detection of alternative and terminal complement pathway proteins in the glomerulus of patients with C3G further highlights the important pathogenetic role of complement.¹⁶⁵

AUTOIMMUNE, GENETIC FORMS AND SECONDARY CAUSES

After the initial hypothesis of a circulating factor in GN patients leading to increased C3 cleavage and subsequent discovery of the C3NeF (see earlier), additional autoantibodies or nephritic factors that prolonged the half-life of the complement alternative or classical C3 or C5 convertases have been detected (Table 36.4).^{166,167} However, C3NeF and other autoantibodies found in C3G/IC-MPNG have also been detected in other renal diseases, such as systemic lupus erythematosus (SLE) and infection-related GN (IRGN), as well as in healthy individuals, rendering the interpretation of their pathogenic role controversial.^{166,168}

Table 36.4 Autoantibodies Associated With C3G/IC-MPGN

Autoantibodies	Incidence	Coexisting With C3Nef	Effect on Complement	Routine Testing
C3NeF	Common		Stabilizes AP C3 convertase	Yes
C4NeF	Rare	Yes	Stabilizes CP C3 and AP and CP C5 convertases	No
C5NeF	Rare	Yes	Stabilizes AP C5 convertase	No
Anti-factor B	Rare	No	Stabilizes AP C3 convertase	No
Anti-C3b	Rare	No	Stabilizes AP C3 convertase	No
Anti-factor H	Rare	Yes	Fluid phase AP regulation	Yes

C3NeF is the most prevalent autoantibody, and its levels can fluctuate during the clinical course with no correlation between titers and disease activity or treatment status.¹⁶⁹ C4NeF has been detected in IC-MPGN and C3G and usually coexists with C3NeF.¹⁷⁰ C5NeF stabilizes the C3 and C5 complement alternative pathway convertases and has been found in patients with C3G and DDD, although it could be C3NeF with terminal pathway dysregulation as opposed to a unique nephritic factor.^{85,171}

Approximately 25% of patients with C3G have variants in complement genes, with genetic etiologies more prevalent in C3G than in DDD and IC-MPGN. To date, mutations have been reported in the following complement genes: *CFH*, *CFHR5*, *CFI*, *MCP*, *C3*, and *CFB*. Familial C3G has been associated with mutations or variants in genes encoding CFH-related proteins (*CFHR*).¹⁷² *CFH* mutations associated with C3G are clustered at the N-terminal C3b binding site of *CFH* and result in loss of function of CFH complement regulatory activity (rather than CFH endothelial binding as in aHUS).^{163,173} Table 36.5 details the pathogenic gene variants, rare variants, and polymorphisms associated with C3G and IC-MPGN.

CLINICAL FEATURES

C3G is a rare disease, with one to two cases per million per year, slightly more of them males.¹⁷⁴ *CFHR5* glomerulopathy is more common in the Cypriot population, with an estimated prevalence of 140 cases per million Cypriot patients.¹⁷⁵ The average age of diagnosis in C3GN is slightly higher than in DDD (30 vs. 19 years, respectively), and in children DDD presents at a younger age with 70% of patients diagnosed before age 16.^{163,176}

The onset of disease in IC-MPGN, DDD, and C3GN involves nonspecific clinical features with significant overlap. Signs and symptoms include nephrotic syndrome, microscopic or macroscopic hematuria, hypertension, and kidney injury.^{153,177} Although rare, rapidly progressive glomerulonephritis has been reported. *CFHR5*-associated C3GN usually presents with persistent microscopic hematuria in childhood and a strong family history of kidney failure in the Cypriot population.^{178,179}

Hypocomplementemia (low C3 levels)¹⁸⁰ is a common feature of presentation. C4 levels are usually normal. However, in a French series C4 levels were reported to be low in 2.4% of patients with IC-MPGN, in 15% with DDD, and in 36% with C3GN in an English cohort.^{163,174} Of the 155 patients in the French series, 46% of those with IC-MPGN had low C3 levels, compared with 59% of those with DDD and 39.6% of those with C3GN.¹⁶³ A series from

England looked at 80 patients and found low C3 levels in 79% with DDD, compared with 48% with C3GN.¹⁷⁴ The data from these series along with other reports suggest that low C3 levels even when coinciding with low C4 levels can suggest a diagnosis of IC-MPGN, DDD, or C3GN but should not be used in isolation.^{163,174} On the other hand, patients can also present with normal C3 levels. In the English series, 43% of all the patients with IC-MPGN, DDD, or C3GN confirmed on biopsy had normal C3 levels at the time of diagnosis.^{163,174} Normal C3 levels have also been noted in patients with *CFHR5*-associated C3GN.¹⁷²

While extrarenal manifestations in C3G patients are rarely seen, acquired partial lipodystrophy (aPL) and visual problems caused by retinal drusen are seen (i.e., C3 deposits, baring a similarity to drusen seen in age-related macular degeneration, consisting of C3 split products and CFH).¹⁸¹

DIFFERENTIAL DIAGNOSIS

C3G and IC-MPGN should be differentiated from other glomerulonephritides resulting from overactivation of the complement system. Low C3 and C4 levels are predominantly seen in the pediatric population in SLE, whereas isolated low C3 levels are typically seen in infection-related GN (IRGN) such as APSGN, C3G, and IC-MPGN. Some patients with C3G and IC-MPGN may, however, also present with low C4 levels (see earlier). Rarely, chronic bacteremia or shunt nephritis can present with low C3 levels and nephritis, where a history of recurrent fevers or ventriculoatrial shunts should increase the index of suspicion for these diagnoses.

A diagnosis of SLE over C3G/IC-MPGN is more likely with the presence of extrarenal manifestations (e.g., arthralgia, rash, neurologic, hepatic, lung or cardiac involvement) and laboratory findings (i.e., increased ANA and dsDNA antibody titers). Lupus is discussed in detail in Chapter 31. Essential cryoglobulinemia is common in adults and rare in children, and it can present with similar systemic findings (dysproteinemias are discussed further in Chapter 35).

Differentiating IRGN/APSGN from C3G/IC-MPGN can be difficult due to their similar clinical presentations and laboratory and histologic findings. Although a biopsy is often not required in IRGN, if carried out, characteristic findings include proliferative GN on light microscopy, C3 and IgG staining on IF, and EM showing subepithelial humplike deposits.¹⁸² However, these findings can also be seen in C3G, making the interpretation difficult. Ultimately the self-limiting, spontaneous resolution of IRGN can be distinguished from the relapsing/remitting, chronic nature of C3G/IC-MPGN through the different clinical evolutions. Further

Table 36.5 Pathogenic Gene Variants, Rare Variants and Polymorphisms Associated With C3G and IC-MPGN.

Gene	Mutation/SNP	Function	Classification	References
CFH	Homo-/compound heterozygous SCRs 1–4 (regulatory domain)	Intact surface binding Reduced C3b binding Loss of CFH cofactor and decay accelerating activity	C3G IC-MPGN	(Levy, Halbwachs-Mecarelli et al. 1986, Vogt, Wyatt et al. 1995, Ault, Schmidt et al. 1997, Dragon-Durey, Fremeaux-Bacchi et al. 2004, Licht, Heinen et al. 2006, Habbig, Mihatsch et al. 2009, Servais, Noel et al. 2012, Bu, Borsa et al. 2016, Iatropoulos, Noris et al. 2016, Iatropoulos, Daina et al. 2018)
CFI	Homozygous Heterozygous	Decreased CFI mediated C3b degradation	C3G IC-MPGN	(Servais, Noel et al. 2012, Bu, Borsa et al. 2016, Iatropoulos, Noris et al. 2016, Iatropoulos, Daina et al. 2018)
C3	Heterozygous	C3mut-resistant to cleavage by C3bBb C3mut convertase-resistant to CFH inactivation C3 binding with CFI or CFH	C3GN IC-MPGN	(Martinez-Barricarte, Heurich et al. 2010, Bu, Borsa et al. 2016, Iatropoulos, Noris et al. 2016, Iatropoulos, Daina et al. 2018)
CFB	Heterozygous/ Homozygous	Alters C3-FB interaction	C3G IC-MPGN	(Iatropoulos, Noris et al. 2016, Iatropoulos, Daina et al. 2018)
Thrombomodulin	Homozygous	Not tested	DDD	(Iatropoulos, Noris et al. 2016, Iatropoulos, Daina et al. 2018)
DGKE	Homozygous Heterozygous—unclear impact	Not complement mediated	MPGN	(Ozaltin, Li et al. 2013, Bu, Borsa et al. 2016)
At risk SNPs (Reviewed in (Noris and Remuzzi 2015))				
MCP/CD46	Rare SNP	Not tested	C3G IC-MPGN	(Servais, Noel et al. 2012)
CFH	Rare SNP e.g., Y402H (SCR 7)	Not tested	DDD	(Hageman, Anderson et al. 2005, Abrera-Abeleda, Nishimura et al. 2006, Abrera-Abeleda, Nishimura et al. 2011)
CFHR5	Rare SNP	Not tested	DDD	(Abrera-Abeleda, Nishimura et al. 2006, Abrera-Abeleda, Nishimura et al. 2011, Bu, Borsa et al. 2016)
C3	Rare SNP	Not tested	DDD	(Smith, Alexander et al. 2007, Abrera-Abeleda, Nishimura et al. 2011)
CFHR Fusion Proteins (Reviewed in (Smith, Appel et al. 2019))				
CFHR3-1	CNV CFHR3-1 hybrid gene	Greater degree of CFHR-mediated deregulation	C3GN	(Malik, Lavin et al. 2012, Goicoechea de Jorge, Caesar et al. 2013)
CFHR2-5	CFHR2-5 hybrid gene	Stabilizes C3 convertase, reduced CFH-mediated decay	DDD	(Chen, Wiesener et al. 2014)
CFHR5-CFHR5	CNV Duplication in CFHR5 exons 2/3	Greater degree of CFHR-mediated deregulation	C3G	(Gale, de Jorge et al. 2010, Goicoechea de Jorge, Caesar et al. 2013)
CFHR1-CFHR1	Internal duplication	Greater degree of CFHR-mediated deregulation	C3G	(Tortajada, Yebenes et al. 2013)
CFHR5-2	CFHR5-2 hybrid gene	Greater degree of CFHR-mediated deregulation	C3GN	(Smith, Appel et al. 2019)
CFHR1-5	CFHR1-2 hybrid gene	Greater degree of CFHR-mediated deregulation	C3G	(Togarsimalemath, Sethi et al. 2017)

CFHR, Complement factor H-related (CFHR) proteins; IC-MPGN, immune complex membranoproliferative glomerulonephritis.

investigations for DDD or C3GN should be considered in the presence of persistent low C3 levels or macroscopic hematuria and/or proteinuria for longer than 2 to 3 months.¹⁸³ Elevated ASO titers can be seen in APSGN but also in C3G.¹⁸⁴

DIAGNOSIS AND DIAGNOSTIC WORKUP INCLUDING THE ROLE OF A KIDNEY BIOPSY

Clinical presentations of IC-MPGN, DDD, and C3GN are varied and can be difficult to distinguish from other

glomerular disorders. They can present as nephrotic range proteinuria, microscopic or macroscopic hematuria, or acute kidney injury. A diagnosis of C3G should be considered in any patient presenting with glomerulonephritis. A diagnostic workup for C3G/IC-MPGN is detailed in [Table 36.6](#).

During the history taking process, attention should be placed on identifying any secondary causes of glomerulonephritis including recent infections and autoimmune illnesses (see [Table 36.3](#)). A family history of glomerulonephritis and previous episodes of hematuria and proteinuria would point toward a complement-mediated form. After a thorough history and examination, the laboratory workup should include an initial evaluation for secondary causes, as well as complement dysregulation (see [Table 36.3](#)). Patients older than 50 years of age should be screened for paraproteinemia and referred to hematology for consideration of clone-directed chemotherapies.^{185,186} Ultimately, the diagnosis can only be made with a kidney biopsy including light microscopy, immunofluorescence, and electron microscopy. As noted earlier, C3G is characterized by the presence of C3 only or C3 dominance on immunofluorescence examination. Dense deposit disease (DDD) is highlighted by dense intramembranous deposition along the glomerular basement membrane by electron microscopy (EM). C3GN can also show electron-dense deposits, but in

contrast more in the mesangium and without the ribbon formation. Immune complex-associated MPGN (IC-MPGN) is characterized by a membranoproliferative light microscopy pattern with immunoglobulin dominance on immunofluorescence examination.

If the initial laboratory and histopathologic findings are suggestive of overactivation of the complement alternative pathway (negative screen for secondary causes and C3 staining on biopsy), then a full complement workup should be carried out, involving a hemolytic assay of the classical and alternative pathway (CH50 and AP50), genetic screening (*C3*, *CFH*, *CFI*, *CFB*, *MCP*, *CFHR5* in patients with Cypriot origin), and measurement of autoantibodies (C3NeF, C5NeF, anti-CFH, anti-CFB, and anti-C3b).¹⁵⁷ Functional assays have yet to be validated. C3G is thought to be a disease of (mainly fluid phase) C3 dysregulation, but some patients show increased levels of soluble C5b-9, indicating terminal pathway activation. This is an interesting finding because those patients might respond to therapeutic C5 inhibition.¹⁸⁷ Of note, a machine learning-based analysis of C3G patients identified C5b-9 levels as predictors for disease severity and patient outcome.¹⁸⁸

MANAGEMENT

There are no evidence-based treatment recommendations in IC-MPGN or C3G due to the paucity of trial data, and instead current management is based on expert opinion, retrospective reviews, and smaller case studies ([Fig. 36.5](#)).⁸⁸

Supportive therapy

The optimization of blood pressure and proteinuria control is important in all patients with IC-MPGN or C3G. An increased level of proteinuria has been associated with an increased risk of developing kidney failure.¹⁸⁹ Patients should receive renin-angiotensin-aldosterone system (RAAS) blockade with angiotensin-converting enzyme inhibitors (ACEIs), angiotensin II receptor antagonists (ARBs), or newer agents once approved for pediatric patients. Their antiproteinuric and antihypertensive effect has been associated with better renal survival in C3G patients.^{189,190} Optimal nutrition including a low-salt diet and lipid control is also recommended.

Immunosuppressive therapy

If proteinuria is >500 mg/24 hours or there is moderate inflammation on kidney biopsy, the use of oral prednisone is recommended. Long-term low-dose prednisone has shown improvement in proteinuria and renal function, although data suggest only a subgroup of patients with IC-MPGN will respond, possibly only partially. Benefit has also been described in C3G patients.

A recommended dosage would involve an induction phase (prednisone given at 60 mg/m² (maximal 60 mg/day) for 4 weeks and then 40 mg/m² on alternate days for another 4 weeks), followed by a maintenance phase (prednisone tapered for a total of 6–12 months). If proteinuria is severe (>2000 mg/24 hours), there is intense inflammation on biopsy, or impaired renal function occurs, consideration of pulses of intravenous methylprednisone (1000 mg/1.73 m² repeated 3 times) is suggested.¹⁹¹

The use of mycophenolate mofetil (MMF) either alone or in addition to prednisone has shown promising results

Table 36.6 Diagnostic Workup for IC-MPGN and C3G

History	Family history of hematuria, glomerulonephritis, and kidney failure Gross hematuria (if yes, concomitant or 2–3 weeks subsequent to infection; number of episodes) Previous urinalysis results? Infection (especially upper respiratory tract) in the preceding 2–3 weeks
Clinical examination	Signs of nephrotic syndrome (peripheral and periorbital edema, weight increase) Blood pressure Presence of frothy urine, reduction in urine output Extrarenal features: partial lipodystrophy, retinal drusen
Laboratory workup	Urinalysis, 24-hour proteinuria, microhematuria Complete blood count, urea, creatinine, protein, IgG, IgA, and IgM Serum C3, C4 Antinuclear antibody, antidouble stranded DNA antibody; serology for hepatitis C and B virus;
Complement workup	C3 nephritic factor (C3Nef), C4Nef and C5Nef if available Anti-FH antibodies; anti-FB antibodies and anti-C3b antibodies if available Soluble C5b9 and total hemolytic complement (CH50); FH, FI, FB levels if available Genetic screening (next-generation sequencing): C3, CFH, CFI, CFB, MCP, CFHR5 in patients with Cypriot origin

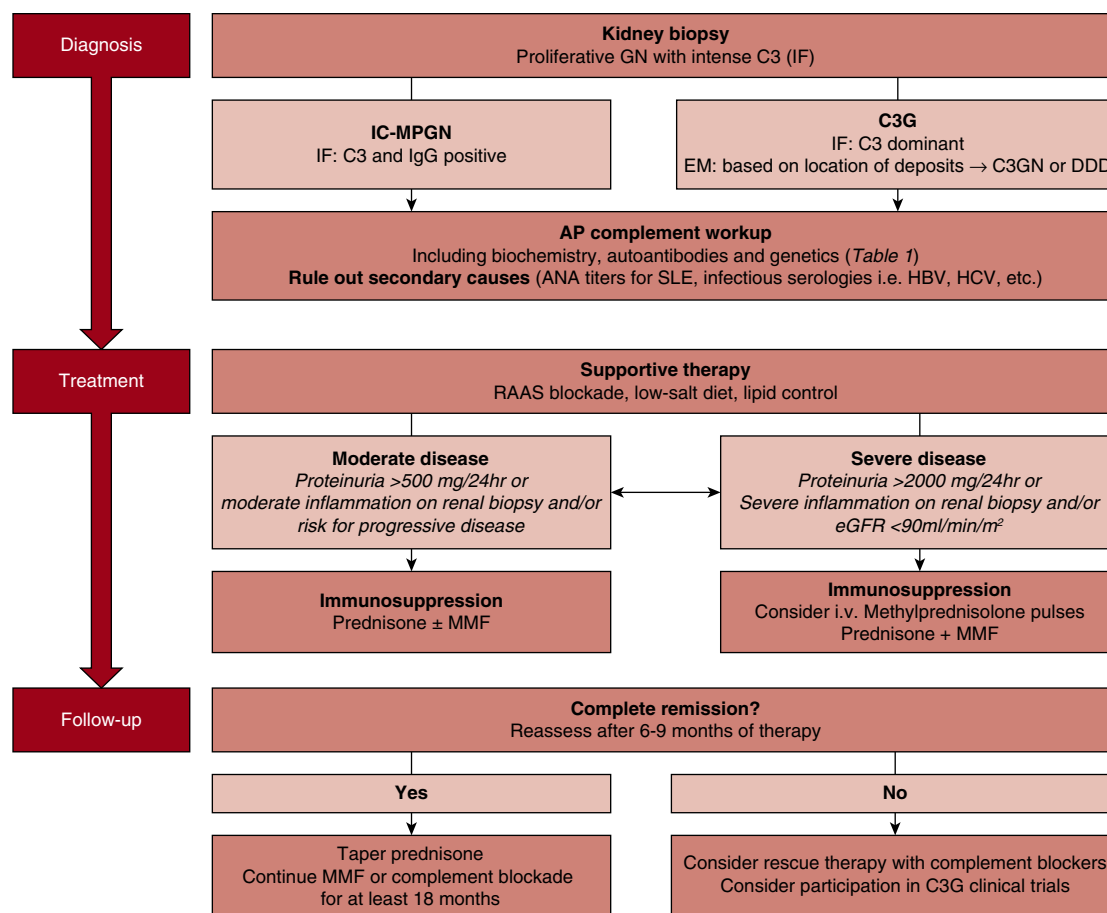


Fig. 36.5 Algorithm for Diagnosis, treatment and follow-up in IC-MPGN and C3G patients. Complement blockers targeting CFB (C3 convertase) and C3 are now available for the treatment of C3G and/or IC-MPGN patients >12 years including patients with post-transplant disease recurrence. Treatment algorithms or protocols—other than the ones used in the clinical trials—are not available, yet. C3 and CFB inhibitors may be used for primary treatment (with or without prednisone/MMF), for treatment escalation in patients not responding to prednisone/MMF, or in patients with severe or progressive disease. While drugs in transplant patients are currently only approved for disease recurrence, preemptive treatment aiming to prevent recurrence should be considered. Also, there are no guidelines defining treatment duration. As for now, therapy duration should be decided on an individual basis and can be discussed with experts.

in idiopathic IC-MPGN. Multiple case series have shown sustained improvement in proteinuria, renal function, and the induction of complete or partial remission in patients with primary MPGN after the addition of MMF.^{192,193} Similar results have been reported in C3G, with patients receiving steroids and MMF having a higher rate of remission and lower chance of kidney failure compared with other immunosuppressive therapy, C5 inhibition, or conservative treatment.¹⁹⁰ These findings have been mirrored in patients with genetic complement abnormalities or autoantibodies to complement components.¹⁹⁰ A Turkish study looked at the effect of MMF treatment on disease progression in 60 children with C3G with and without mutations in complement-related genes and found no difference in kidney survival between the groups.¹⁹⁴

In our experience, as recommended by the KDIGO expert guidance, the use of MMF in IC-MPGN and C3G with severe proteinuria should be considered and continued for a total of 18 to 24 months.⁸⁸

In steroid-resistant MPGN, calcineurin inhibitors (tacrolimus and cyclosporin) have been beneficial in inducing long-term reduction of proteinuria and stable renal function, with low rates of recurrence after treatment cessation both in pediatric and adult patients.¹⁹⁵⁻¹⁹⁸ However,

some data show no improvement in renal survival with calcineurin inhibitor administration.¹⁹⁹ In our experience, calcineurin inhibition shows a benefit in some patients, inducing remission in both IC-MPGN and C3G, especially in patients with persistent and significant proteinuria despite 6 to 9 months of immunosuppressive therapy.

Lowering C3NeF levels using B cell-depleting agents such as rituximab has shown conflicting results. There are case reports in patients with IC-MPGN where rituximab has induced partial or complete remission.²⁰⁰⁻²⁰² However, despite a reduction in C3NeF levels, some patients showed no change in proteinuria or renal function and one patient later showed clinical improvement with C5 inhibition.^{202,203} Even in C3NeF-positive patients, the role of rituximab is unclear, with most authors finding no benefit after its administration. Whether this lack of benefit is because C3NeFs are not drivers of disease or whether their levels need to be reduced to zero before any clinical effect is seen is yet to be determined.

Complement-targeting therapy

Complement AP dysregulation in IC-MPGN and C3G may indicate that complement inhibition could be a potential therapeutic target for these glomerular diseases. Ideally, these

medications would target complement at the level of C3, as this is where complement dysregulation occurs in patients with C3G/IC-MPGN. Eculizumab, a monoclonal antibody against C5, only targets the terminal pathway and, as noted earlier, has shown great benefit in patients with aHUS.⁹⁵ In C3G/IC-MPGN, eculizumab seems to work only in a subset of patients. Complement inhibitors blocking at the C3 level are currently (e.g., pegcetacoplan) under clinical investigation. Historically, there are case reports that PLEX induces partial or complete remission and reduces the risk of recurrence in patients with IC-MPGN and C3G.²⁰⁴⁻²⁰⁶ However, in one patient with IC-MPGN, after an initial improvement in renal function and proteinuria with PLEX, there was a subsequent clinical deterioration with seizures, respiratory distress, and anuria. The patient was switched to eculizumab therapy given evidence of complement alternative and terminal pathway overactivation with low C3 levels, elevated sC5b-9, and a deletion for *CFHR1*, which resulted in improved proteinuria, renal function, and clinical status.²⁰⁷

Chronic infusions of fresh frozen plasma (FFP) as maintenance therapy have been shown to stabilize and maintain renal function,²⁰⁸ although plasma infusions are unlikely to be sufficient as induction therapy.²⁰⁹

Terminal pathway blockade with eculizumab (monoclonal anti-C5 antibodies) has shown mixed results. There have been several reports of its successful use in IC-MPGN, C3G, and DDD,^{207,210} with a case series in Italy showing a significant response to eculizumab in 10 of 13 patients including C3G, DDD in native and transplanted kidneys.¹⁸⁷ Better outcomes seem to be associated with higher sC5b-9 levels and shorter disease duration.¹⁸⁷ However, a retrospective study found limited benefit, with only 6 of 11 patients showing a decrease in >30% in eGFR after administration of eculizumab.²¹¹ A French case series showed a global clinical response in patients with a more rapidly progressive course and more intense glomerular extracapillary proliferation, suggesting that eculizumab targets mainly glomerular inflammation secondary to AP complement dysregulation.²¹²

The EAGLE study²¹³ assessed the use of eculizumab in IC-MPGN and C3G patients with normal renal function, severe proteinuria (>3500 mg/24 hours), and significantly elevated sC5b9 (>1000 ng/mL). The results were disappointing despite effective terminal complement pathway blockade, with only three of seven patients achieving partial remission of proteinuria. However, the assessment of nonresponse to eculizumab in this setting must take into account that eculizumab may be lost through the urine in patients with significant proteinuria, resulting in incomplete C5 inhibition, which might require more frequent dosing.²¹⁴ Overall, in patients with intense glomerular inflammation, elevated sC5b9, or a mixed aHUS/C3G picture, a trial of C5 inhibition seems a reasonable step as a rescue therapy when immunosuppression has failed.

Avacopan, an oral C5a inhibitor, was investigated in a randomized, double-blind, placebo-controlled trial (ACCOLADE) and showed some attenuation in C3G progression by measuring a disease chronicity score in kidney biopsies of patients with C3G.²¹⁵

With the pathophysiology in mind, complement inhibition at the level of the AP C3 convertase has appropriately received serious consideration. Several trials of a C3 inhibitor are under way, a factor B inhibitor and a factor D inhibitor. Phase 2 results of pegcetacoplan, a targeted C3 and C3b inhibitor, were reported in 2023 showing an acceptable safety profile and mean proteinuria reduction of 50% over 48 weeks, stable kidney

function over time, normalization of albumin, and a rise in C3 with a concomitant decrease of soluble C5b-9.²¹⁶

In the Phase 3 VALIANT study, pegcetacoplan demonstrated a 68% reduction in proteinuria and stabilization of kidney function, including patients post-transplantation (unpublished, <https://investors.apellis.com/news-releases/news-release-details/fda-approves-apellis-empavelir-pegcetacoplan-first-c3g-and>). Pegcetacoplan was approved by the FDA for the treatment of C3G and primary IC-MPGN for patients 12 and older in July 2025. The latter indicates that targeting C3 has an impact on the underlying disease mechanism, as C5 inhibition does not affect C3 levels.²¹⁴ Although promising, treatment protocols and long-term management have not been established and need to be considered at an individual patient level. A suggested proposal is made in Fig. 36.5.

Iptacopan, an oral factor B inhibitor, is currently being evaluated in a double-blinded randomized placebo-controlled trial (APPEAR-C3G) in patients with C3 glomerulopathy (C3G).²¹⁷ The phase 2 study, iptacopan, was associated with a mean increase in eGFR of 3.1 mL/min per 1.73 m² from baseline to 12 weeks in patients with native C3G, a significant 45% reduction in proteinuria, and a significant reduction in the histologic C3 deposit score on follow-up kidney biopsy in patients with recurrent C3G post-transplant.²¹⁷ Preliminary results from the APPEAR-C3G study showed that the primary endpoint of the Phase III study was met, showing proteinuria reduction at 6 months (35%), which was sustained at 12 months. The composite endpoint (50% proteinuria reduction and less than 15% reduction in eGFR at 12 months) was met in 43.5%.^{217a} The FDA approved iptacopan for the treatment of C3G in adults in March 2025.

Danicopan, an oral factor D inhibitor, failed to achieve optimal and sustained systemic concentrations in two phase 2 trials, and therefore only limited clinical response was observed.²¹⁸ The final results of the phase 3 trials are expected, but targeted treatments for patients with C3G and IC-MPGN seem to be on the horizon.

CLINICAL OUTCOMES

There are little long-term data on outcomes with the updated classification for MPGN, DDD, and C3GN. Multiple studies have shown that around 50% of patients with MPGN progress to kidney failure within 10 years, irrespective of the type.¹⁶¹ In a cohort of 80 patients with DDD and C3GN, there was progression to kidney failure in 47% and 23% of patients, respectively, with a median follow-up of 28 months and no difference in cumulative kidney survival, although they did report that age older than 16 years and crescents on biopsy were independent predictors of disease progression.¹⁷⁴

Servais et al¹⁶³ reported 10-year kidney survival of 63.5% with no difference among MPGN, DDD, and C3GN groups in a cohort of 134 patients. Both studies report that that eGFR at diagnosis and age at presentation were inversely correlated with long-term renal survival.¹⁷⁴ Children are reported to have better outcomes than adult, with a cohort of 165 children with IC-MPGN and C3G showing preserved renal function at 4 years, and at 10 years, 80% of children did not meet the composite outcome of eGFR <30 mL/min/1.73 m², 50% eGFR reduction, or need for kidney replacement therapy.²¹⁹

TRANSPLANTATION

The disease recurrence rate for MPGN and DDD in transplants is high, with ESPN/ERA-EDTA registry data

showing the 5-year posttransplant risk of graft loss at 32.4%.²²⁰ Graft loss has been seen in 70% of DDD and 30% of C3GN patients and, on average, at around 43 months post transplant.^{221,222} Servais et al¹⁶³ reported recurrence rates of 43% in MPGN, 55% in DDD, and 60% in C3GN. Children have an increased risk of disease recurrence, although the reason for this is unclear.²²³ Zand reported outcomes in 21 transplant patients and interestingly noted three differences in the recurrence group: more females, a higher number carrying the HLA-DRI7-DQ2 haplotype (also seen in type 1 diabetes mellitus), and lower pretransplant C3 levels.²²⁴ Medjeral et al⁷³ reported 10 of 13 transplant patients had recurrence of DDD or C3GN, and this was strongly associated with graft failure.¹⁷⁴ CFHR5-associated C3GN has also been reported to recur in renal transplants.

FUTURE PERSPECTIVES

Further work is needed to better classify patients at risk for kidney failure in C3G and IC-MPGN. The pathophysiology is still not well understood, especially the continuum of a more inflammatory phenotype (IC-MPGN) first and then a more prominent C3G pattern in later biopsies. Can immunosuppressive therapy mitigate this injury pattern? Furthermore, C3 targeting therapies are being evaluated, and so far they show promising results for patients with a devastating disease: high rates of kidney failure and even higher rates of recurrence post transplant.

ACUTE POSTSTREPTOCOCCAL GLOMERULONEPHRITIS

Acute poststreptococcal glomerulonephritis (APSGN) is the prototypical postinfectious (often synpharyngitic) immune complex glomerulonephritis. Typically, it occurs after either a skin (2–6 weeks) or throat infection (1–2 weeks) caused by M-type substrains of group A streptococcus (*Streptococcus pyogenes*) and occasionally by *Streptococcus* group C or G.

PATHOGENESIS

Several nephritogenic antigens (NAPlr, SPEB, streptokinase) have been described. They are produced as exotoxins, and upon entering circulation they deposit in the glomerular basement membrane. Immunoglobulins will then bind those antigens and form immune complexes activating the alternative pathway of the complement system and leading to C3 deposition within the glomerulus. Due to the massive amount of C3 activation/deposition, patients will present with low plasma C3 levels. Activation of the cellular immune system will result in lymphocyte infiltration, IL-8 production, and secondary neutrophil infiltration, resulting in nephritis. Autoimmune phenomena have also been reported in patients with APSGN including rheumatoid factor, anti-IgG deposits in kidney biopsies, anti-DNA, anti-C1q antibodies, and antineutrophil cytoplasmic antibodies.²²⁵

One such nephritogenic protein is nephritis-associated plasmin receptor (NAPlr). It has been shown to deposit on the endothelial side of the GBM and mesangial matrix and bind plasmin.²²⁶ Plasmin can then degrade extracellular matrix proteins such as fibronectin and laminin and cause activation of monocytes, macrophages, and neutrophils, leading to glomerular inflammation. Furthermore, NAPlr has also been shown to activate the alternative complement

pathway by converting C3 into C3b.²²⁶ The search for other nephritogenic streptococcal proteins has also identified streptococcal pyrogenic exotoxin B (SPEB), a serine protease that localizes to glomeruli in PSGN.²²⁷ Similar to NAPlr, it can bind plasmin to induce endocapillary glomerular inflammation. Interestingly, SPEB has been shown to bind directly to C3.²²⁸ It leads to C3 degradation, though it is unclear if it does so in a way that leads to formation of C3b, as this degradation is associated with reduced opsonization of the surface of Group A *Streptococcus*.²²⁹

EPIDEMIOLOGY

The prevalence of APSGN varies widely and is influenced by geographic, socioeconomic, host, and bacterial factors (for further discussions see [Chapter 34](#)). Cases have been declining in developed areas, but it remains a significant problem in less developed countries with an annual incidence of 450,000 cases.²³⁰ Epidemics have also been described.²³¹

Children aged 4 to 14 years are most affected, but APSGN can also occur in young adults, both with a male predominance. Older adults, often diabetic or immunocompromised, may also develop postinfectious GN, more often from visceral sites of infection, and have a poor prognosis with frequent progression to end-stage kidney disease.²³² Greater latency between infection and nephritis is reported after skin infection (3–6 weeks) than respiratory infection (1–2 weeks).

CLINICAL PRESENTATION

APSGN presentation varies from asymptomatic hematuria to acute nephritic syndrome, nephritic-nephrotic syndrome, and rarely rapid progressive crescentic glomerulonephritis. Asymptomatic hematuria can be accompanied by arterial hypertension. In a prospective study of household members of index cases, asymptomatic patients were up to five times more frequent than symptomatic patients.²³³ APSGN patients typically present with an acute nephritic picture: hematuria, edema, hypertension, and moderate proteinuria. Glomerular hematuria can be identified by urine microscopy showing red cell casts and dysmorphic RBCs. Gross hematuria (cola-colored urine) occurs in one-third of the patients. Edema is evident in 90% of symptomatic patients, hypertension in 60% to 80%, with half requiring antihypertensive therapy. Decreased urine output is reported in half of all symptomatic children, but the need for dialysis is rare. A rapid progressive course occurs in <1% of the cases and, when present, is due to crescentic glomerulonephritis. Other nonspecific symptoms include general malaise, nausea, weakness, headache, and dull lumbar pain. Patients older than 65 years were more likely to have hypertension and congestive heart failure.²³⁴ Macroscopic hematuria usually resolves within a few days, but microscopic hematuria can persist for months/years and might exacerbate during intercurrent illness or strenuous exercise. Edema and hypertension typically resolve within a few weeks. Persistent high levels of proteinuria are a risk factor for progression to CKD. The acute nephritic syndrome stems from the inflammatory reaction in the glomeruli. Sodium and water retention causes expansion of extracellular volume contributing to edema and hypertension.

DIAGNOSIS

Reduction in serum complement C3 levels is a hallmark feature of APSGN. Classical and lectin pathway might also be activated, and low C4 levels have also been described.²³⁵ Complement levels usually return to normal within 1 to 3 months.¹⁸³ IgG and IgM serum levels are elevated in the majority, and elevated antistreptococcal antibody titers (antistreptolysin O and anti-DNAse B) are the usual laboratory confirmation of a preceding streptococcal infection, especially as positive cultures are obtained in only 20% to 25% of cases outside of epidemics.²³⁶ Other immunologic markers as described earlier can be seen in patients with APSGN. Renal biopsy is rarely done in “typical” patients with APSGN, but it is indicated in patients with rapid progression of kidney failure, persistent worsening proteinuria, and persistently low C3 (beyond 3 months) to investigate different diagnoses, such as C3G. Light microscopy shows a diffuse proliferative glomerulonephritis in the early phase with neutrophil infiltrates and later mainly showing mesangial proliferation. Immunofluorescence in the first few weeks will show mainly fine granular C3 and IgG deposits in the glomerular capillary walls and mesangium (“starry sky”), whereas later predominant C3 in the mesangium is described.²³² Immunoglobulin A may be the dominant immunoglobulin deposited, particularly in older patients and patients with diabetes.²³² The characteristic findings on electron microscopy are large subepithelial electron dense deposits also called “humps.” Subepithelial humps are highly suggestive of postinfectious GN but may also be seen in C3G.²³²

Although APSGN is the most common reason for a nephritic syndrome in a child, as a preceding infection it may have been subclinical or so mild as not to be recalled, when there is no compelling history it is important to rule out other causes of acute nephritic syndrome. The most useful test is complement C3, to differentiate between hypocomplementemic and nonhypocomplementemic GN.

TREATMENT

If streptococcal infection is present at the time of diagnosis, it should be treated with penicillin or another effective antibiotic.¹⁵² In epidemic situations and in high-risk settings, preemptive antibiotic treatment has been shown to decrease the number of APSGN cases. Early administration is reported to prevent or ameliorate the nephritis.²³⁷ If active infection is not present, antibiotic treatment is generally not provided. Household members should be screened for a streptococcal infection. Prophylactic antibiotic treatment is indicated in epidemic situations and may be considered in household members of actively infected index cases in nonepidemic conditions.¹⁵²

Treatment of the glomerulonephritis is symptomatic and includes management of edema and hypertension. With an RPGN picture and kidney biopsy showing crescents, many clinicians will provide pulse methylprednisolone, followed by a course of high-dose oral steroids. No randomized trials exist for this or any other treatments in APSGN.

Overall, the long-term prognosis is quite good. However, some studies have shown that long-term follow-up and screening for hypertension and microalbuminuria is

necessary as 10% to 20% develop sequelae within 10 to 20 years of presentation.^{238,239} This might be explained, as in other diseases with substantial kidney injury, by glomerular hyperfiltration, which in the long term leads to glomerular hypertension and mechanical stress at the filtration barrier.²⁴⁰

In summary, complement-mediated diseases are an exciting field in nephrology. Although there is still much to discover in terms of their disease mechanism, patients with these diseases have a brighter future ahead as several complement inhibitors are available or being tested in clinical trials.

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